PHYSICS • GRADE 9

Detailed study notes

Kinematics, astronomy, dynamics, conservation laws, oscillations and waves, atomic and nuclear physics

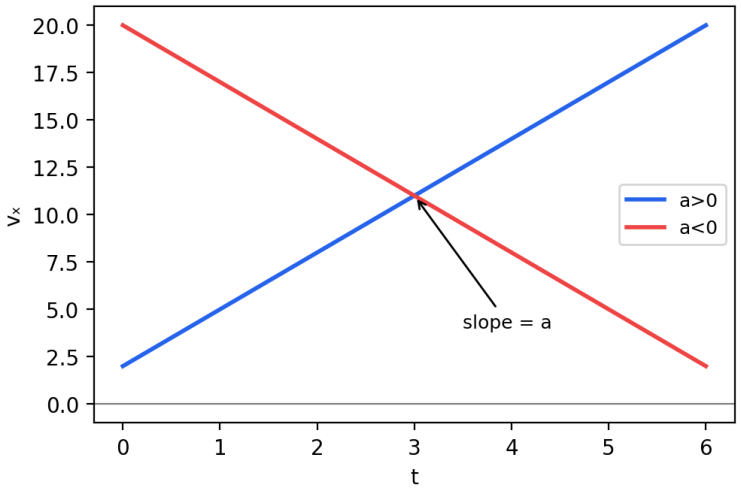
Based on the textbook: N. A. Zakirova & R. R. Ashirov. Physics. Grade 9. — Nur-Sultan: Arman-PV, 2019.

Contents

Contents 2
Chapter 1. Fundamentals of kinematics 4
§ 1. Mechanical motion 4
§ 2. Vectors and operations on them, projection of a vector on coordinate axes 5
§ 3. Uniformly accelerated rectilinear motion, acceleration 6
§ 4. Velocity and displacement in uniformly accelerated rectilinear motion 7
§ 5. Free fall of bodies, acceleration of free fall 8
§ 6. Curvilinear motion, uniform motion of a particle around a circle. Linear and angular velocity 10
§ 7. Centripetal acceleration 11
Chapter 2. Fundamentals of astronomy 13
§ 8. The starry sky 13
§ 9. The celestial sphere, celestial coordinate systems 14
§ 10. Apparent motion of celestial bodies at various latitudes; local, zone and universal time 15
§ 11. The laws of planetary motion in the Solar System 17
§ 12. Determining distances in astronomy using the parallax method 18
Chapter 3. Fundamentals of dynamics 20
§ 13. Newton's first law, inertial reference frames 20
§ 14. Forces in mechanics 21
§ 15. Newton's second law, mass 22
§ 16. Newton's third law 23
§ 17. The law of universal gravitation 24
§ 18. Weight of a body, weightlessness 25
§ 19. Motion of a body under gravity. Motion of artificial Earth satellites 26
Chapter 4. Conservation laws 28
§ 20. Momentum of a body and impulse of force. Conservation of momentum 28
§ 21. Reactive (jet/rocket) motion 29
§ 22. Mechanical work and energy 30
§ 23. Conservation and transformation of energy 31
Chapter 5. Oscillations and waves 33
§ 24. Oscillatory motion 33
§ 25. Energy transformation during oscillations. The equation of oscillatory motion 34
§ 26. Oscillations of simple and spring pendulums 35
§ 27. Free and forced oscillations, resonance 36
§ 28. Free electromagnetic oscillations 37
§ 29. Wave motion 38
§ 30. Sound, characteristics of sound, acoustic resonance, echo 39
§ 31. Electromagnetic waves. The electromagnetic wave spectrum 40

Chapter 6. Atomic structure, atomic phenomena 42
§ 32. Thermal radiation, light as an electromagnetic wave 42
§ 33. The photoelectric effect and its laws, applications 43
§ 34. The structure of the atom. Rutherford's experiments 44
§ 35. Bohr's quantum postulates, the Bohr model of the hydrogen atom 45
§ 36. Lasers, applications of lasers 46
Chapter 7. The atomic nucleus 48
§ 37. Composition of the atomic nucleus, nuclear forces, binding energy 48
§ 38. Radioactivity, the law of radioactive decay 49
§ 39. Nuclear reactions, nuclear fission 50
§ 40. Thermonuclear reaction, nuclear power engineering 51
§ 41. Elementary particles 52
Chapter 8. Conclusion. The modern physical picture of the world 54
§ 42. Physics and scientific-technological progress 54
§ 43. The physical picture of the world 55

Chapter 1. Fundamentals of kinematics



§ 1. Mechanical motion

- Mechanical motion is the change of a body's position in space relative to other bodies over time.
- A material point (particle) is a body whose shape and size can be neglected in a given problem.
- A reference body, coordinate system and a time-measuring instrument together form a frame of reference; motion is relative (a passenger is at rest relative to the seat but moving relative to the ground).

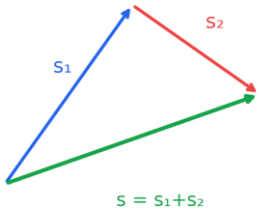
DEFINITIONS

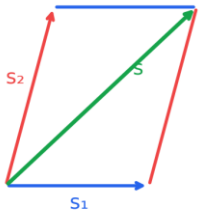
Term	Definition
Kinematics	the branch of mechanics that studies motion without investigating its causes.
Material point	a body whose shape and size can be neglected in the given problem.
Frame of reference	a reference body, its coordinate system, and a clock for measuring time.

FORMULAE AND RELATIONS

Formula	Note
u = u_rel + u_transp	velocity addition theorem: velocity relative to a fixed frame equals the geometric sum of relative and transport (frame) velocity.
s = s_rel + s_transp	displacement addition theorem (analogous to velocities).

DIAGRAM §1





KEY POINT
A geometric sum differs from an algebraic one by accounting for vector direction: if the velocities point the same way, u = u_rel + u_transp; if opposite, u = u_rel − u_transp.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 1, item 2

A ship's speed going downstream is 21 km/h, and upstream 17 km/h. Find the ship's speed in still water.

Given	SI	Solution
u_1 (downstream) = 21 km/h u_2 (upstream) = 17 km/h	*u_1 = u_ship + u_current;* *u_2 = u_ship − u_current*	u_ship = (u_1 + u_2)/2 u_ship = (21 + 17)/2 u_ship = 19 km/h

Answer: The ship's speed in still water is 19 km/h.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 2. Vectors and operations on them, projection of a vector on coordinate axes

- The sum of two vectors is found by the triangle rule (joining the end of the first to the start of the second) or the parallelogram rule (vectors from one point, completing a parallelogram).
- The sum of several vectors is found by the polygon rule; the difference of vectors equals the sum of the first and the vector opposite to the second.
- A vector's projection on an axis is the segment between the projections of its start and end; it is positive if motion along the axis matches its direction, negative otherwise.

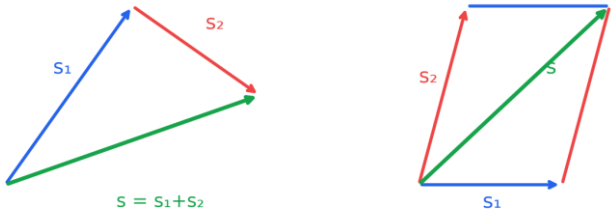
DEFINITIONS

Term	Definition
Triangle rule	to add vectors, join the end of the first to the start of the second; the vector from the first's start to the second's end is their sum.
Vector projection	a segment connecting the projection of a vector's start point to the projection of its end point.
Collinear vectors	vectors lying on the same line or on parallel lines.

FORMULAE AND RELATIONS

Formula	Note
$s = \sqrt{(s_1^2 + s_2^2)}$	magnitude of the sum of two perpendicular vectors (Pythagorean theorem).
$s_x = \upsilon \cdot \cos\alpha;\ s_\gamma = \upsilon \cdot \sin\alpha$	resolving a vector into components along the 0x and 0y axes.

DIAGRAM §2



KEY POINT
A vector's projection is a scalar (can be negative), while its component is a vector; the magnitude from projections uses the Pythagorean theorem: $\upsilon = \sqrt{(\upsilon_x^2+\upsilon_\gamma^2)}$.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 2, item 3

A body moved from a point with coordinates $x_1 = 0$, $y_1 = 2$ m to a point with coordinates $x_2 = 4$ m, $y_2 = -1$ m. Find the magnitude of the displacement and its projections onto the axes.

Given	SI	Solution
$x_1 = 0$ $y_1 = 2$ m $x_2 = 4$ m $y_2 = -1$ m	*All values are in SI*	$s_x = x_2 - x_1 = 4 - 0 = 4$ m $s_\gamma = y_2 - y_1 = -1 - 2 = -3$ m $s = \sqrt{(s_x^2+s_\gamma^2)} = \sqrt{(16+9)} = 5$ m

Answer: $s = 5$ m; $s_x = 4$ m; $s_\gamma = -3$ m.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 3. Uniformly accelerated rectilinear motion, acceleration

- Uniformly variable motion is motion where the velocity changes by the same amount over any equal small time intervals; instantaneous velocity is the velocity at a given moment.
- Acceleration characterizes how fast velocity changes: $a = (\upsilon - \upsilon_0)/\Delta t$. If acceleration is co-directed with velocity, motion is uniformly accelerated; if opposite, uniformly decelerated.
- On a velocity-time graph, acceleration equals the slope of the line; the area under the graph numerically equals the distance travelled.

DEFINITIONS

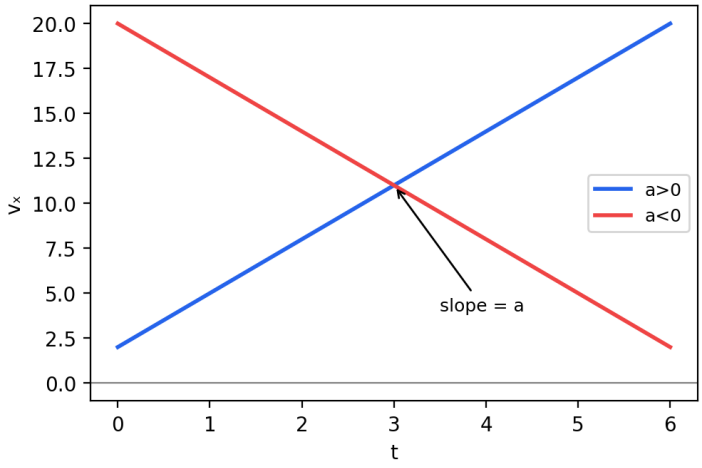
Term	Definition
Instantaneous velocity	the velocity at a given moment; the limit of displacement over time interval as the interval tends to zero.

Term	Definition
Acceleration	a physical quantity characterizing how fast a body's velocity changes.

FORMULAE AND RELATIONS

Formula	Note
$a = (\upsilon - \upsilon_0)/\Delta t$	definition of acceleration via change of velocity.
$[a] = 1\ m/s^2$	the SI unit of acceleration.

DIAGRAM §3



KEY POINT
The steeper the velocity-time graph, the greater the acceleration; where the graph crosses the time axis marks when the body stops.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 3, item 1

A car moving uniformly covers a third of the distance at 20 m/s, and the rest at 36 km/h. Find the average speed over the whole distance.

Given	SI	Solution
$\upsilon_1 = 20$ m/s $\upsilon_2 = 36$ km/h	*$\upsilon_2 = 10$ m/s*	$\upsilon_avg = s/t = s/(s/(3\upsilon_1) + 2s/(3\upsilon_2))$ $\upsilon_avg = 3\upsilon_1\upsilon_2/(\upsilon_2 + 2\upsilon_1)$ $\upsilon_avg = 3 \cdot 20 \cdot 10/(10+40) = 12$ m/s

Answer: The average speed is 12 m/s.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 4. Velocity and displacement in uniformly accelerated rectilinear motion

- Velocity in uniformly variable motion: $\upsilon_x = \upsilon_{0x} + a_x t$; the signs of the projections determine the type of motion (same signs — accelerated, different — decelerated).
- Displacement equals the area under the velocity graph: $s_x = \upsilon_{0x} t + a_x t^2/2$, and also $s_x = (\upsilon_x^2 - \upsilon_{0x}^2)/(2a_x)$, which doesn't involve time.
- The equation of motion $x(t) = x_0 + \upsilon_{0x} t + a_x t^2/2$ gives the body's coordinate at any moment; for uniform motion it reduces to $x = x_0 + \upsilon_x t$.

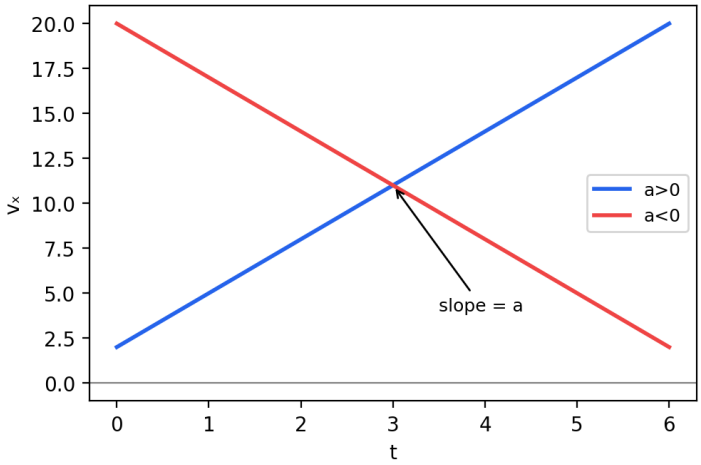
DEFINITIONS

Term	Definition
Equation (law) of motion	the coordinate-vs-time function x(t) that gives the body's location at any moment.
Meeting condition	the equality of the bodies' coordinates $x_1 = x_2$ at the moment they meet.

FORMULAE AND RELATIONS

Formula	Note
$s_x = \upsilon_{0x}t + a_x t^2/2$	displacement in uniformly variable motion.
$s_x = (\upsilon_x^2 - \upsilon_{0x}^2)/(2a_x)$	displacement without time.
$x(t) = x_0 + \upsilon_{0x}t + a_x t^2/2$	equation of motion (coordinate vs time).

DIAGRAM §4



KEY POINT
Kinematics problem-solving algorithm: record data and convert to SI → sketch the body and vectors → choose an axis → write formulas in projections → solve the equation → state the answer.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

Find the time it takes a motorist, starting with acceleration 10 m/s², to catch up with a cyclist moving at 10 m/s. At the moment the car starts, the distance between them was 240 m.

Given	SI	Solution
$a_1 = 10$ m/s² $\upsilon_{10} = 0$ $\upsilon_2 = 10$ m/s $l = 240$ m	*All values are in SI*	$x_1 = a_1t^2/2$; $x_2 = l + \upsilon_2 t$ $a_1t^2/2 = l + \upsilon_2 t \rightarrow 5t^2 = 240 + 10t$ $t_1 = 8$ c ($t_2 = -6$ c — not valid); $x = 320$ m

Answer: $\Delta t = 8$ s; the meeting point is $x = 320$ m.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 5. Free fall of bodies, acceleration of free fall

- Galileo experimentally disproved Aristotle's claim that heavier bodies fall faster than lighter ones: in a vacuum, all bodies fall with the same acceleration g.
- The acceleration of free fall on Earth averages g ≈ 9.81 m/s² (9.83 m/s² at the poles, 9.78 m/s² at the equator) and points vertically down; on other planets g differs.
- Free fall is a special case of uniformly variable motion; formulas for a body thrown vertically, at an angle, or horizontally follow from the general kinematic equations with a = g.

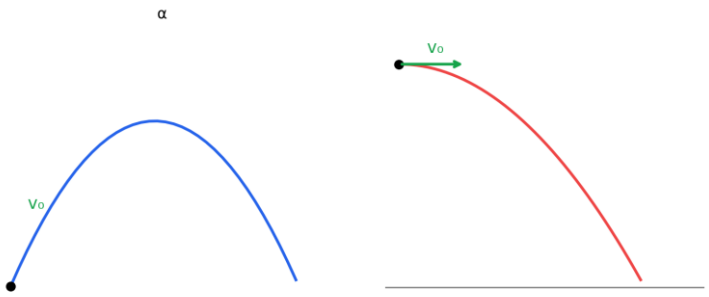
DEFINITIONS

Term	Definition
Free fall	the motion of bodies in a vacuum under the action of gravity alone.
Acceleration of free fall g	the acceleration with which all bodies fall in a vacuum regardless of mass; g ≈ 9.81 m/s² on Earth.

FORMULAE AND RELATIONS

Formula	Note
$h = \upsilon_0 t \pm gt^2/2$	displacement (height) in free fall with initial speed.
$\upsilon_\gamma = \upsilon 0 \sin\alpha - gt$; $h = (\upsilon 0 \sin\alpha)t - gt^2/2$	a body thrown at angle α to the horizontal.
h1:h2:h3 = 1:3:5	the ratio of displacements in successive equal time intervals starting from rest (Galileo's law).

DIAGRAM §5



KEY POINT
The signs of the vertical velocity and displacement projections are positive while the body moves up and negative while it falls (with the 0y axis pointing up).

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

A body is thrown vertically upward at 20 m/s. Find the time interval after which the body is at a height of 15 m above the throw point.

Given	SI	Solution
υ_0 = 20 m/s h = 15 m	*g = 9.8 m/s², 0y axis upward*	$y(t) = \upsilon 0t - gt^2/2$ $15 = 20t - 5t^2$ t_1 = 1 s (ascent); t_2 = 3 s (descent)

Answer: t_1 = 1 s; t_2 = 3 s.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 6. Curvilinear motion, uniform motion of a particle around a circle. Linear and angular velocity

- Curvilinear motion is motion whose trajectory is a curve; any curve can be represented as a combination of circular arcs.
- Period T is the time for one full revolution; frequency ν = 1/T is the number of revolutions per unit time (unit: hertz).
- Angular displacement φ is measured in radians; angular velocity ω = φ/t; linear speed relates to angular speed by υ = ωR.

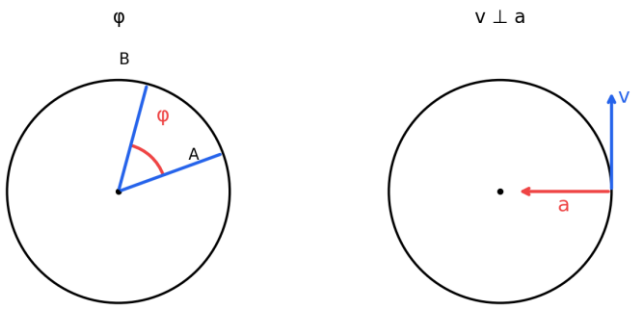
DEFINITIONS

Term	Definition
Period T	the time for one complete revolution.
Frequency ν	the number of revolutions per unit time; ν = 1/T.
Angular velocity ω	the ratio of angular displacement to the time interval over which it occurs.

FORMULAE AND RELATIONS

Formula	Note
υ = 2πR/T = 2πRν	linear speed via period and frequency.
ω = 2π/T = 2πν	angular speed via period and frequency.
υ = ωR	relation between linear and angular speed.

DIAGRAM §6



KEY POINT
The instantaneous velocity in circular motion points along the tangent to the path at that point; since direction constantly changes, circular motion is not uniform in the vector sense.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 6, item 2

A particle moving uniformly around a circle of radius 50 cm covers half its circumference in 10 s. Find the particle's linear speed.

Given	SI	Solution
R = 50 cm t = 10 s	*R = 0.5 m*	l = πR = 3.14·0.5 ≈ 1.57 m υ = l/t υ = 1.57/10 ≈ 0.16 m/s

Given	SI	Solution
l = πR (half the circumference)		

Answer: υ ≈ 0.16 m/s.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 7. Centripetal acceleration

- In circular motion the direction of velocity changes, so the body has acceleration; the acceleration vector points toward the centre of the circle — called centripetal acceleration.
- The magnitude of centripetal acceleration is a = υ^2/R; it can also be expressed via period, frequency, and angular velocity.
- Centripetal acceleration is perpendicular to velocity and therefore changes only its direction, not its magnitude.

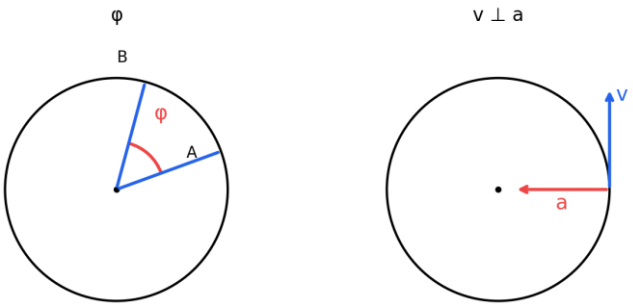
DEFINITIONS

Term	Definition
Centripetal acceleration	the acceleration of a body moving in a circle, directed toward the centre.

FORMULAE AND RELATIONS

Formula	Note
$a = \upsilon^2/R$	centripetal acceleration via linear speed.
$a = 4\pi^2R/T^2 = 4\pi^2\nu^2R$	via period and frequency.
$a = \omega^2R = \omega\cdot\upsilon$	via angular speed.

DIAGRAM §7



KEY POINT
If acceleration is perpendicular to velocity, the body moves in a circle; centripetal acceleration is not a constant vector (its direction changes), though its magnitude is constant for uniform rotation.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

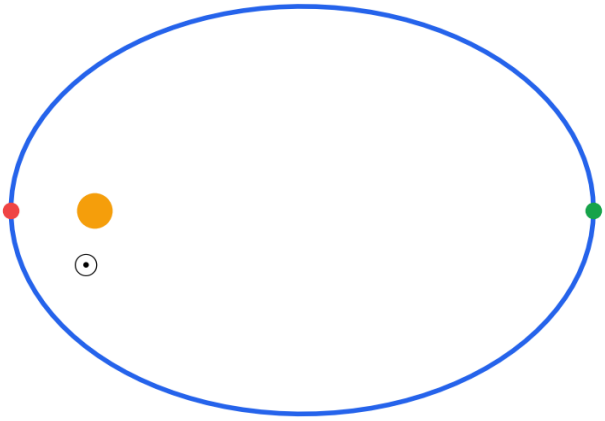
A body's linear speed in circular motion doubles while the radius decreases threefold. Find by how much the acceleration increases.

Given	SI	Solution
$\upsilon_2 = 2\upsilon_1$ $R_2 = R_1/3$	$a_1 = \upsilon_1^2/R_1$; $a_2 = \upsilon_2^2/R_2$	$a_2 = (2\upsilon_1)^2/(R_1/3) = 4\upsilon_1^2 \cdot 3/R_1 = 12 \cdot (\upsilon_1^2/R_1)$ $a_2 = 12a_1$

Answer: The acceleration increases 12-fold.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 2. Fundamentals of astronomy



§ 8. The starry sky

- The Universe is the entire existing material world, consisting of planets, stars, interstellar matter and cosmic rays; the Sun lies about 2.8×10^{4} light-years from the centre of the Milky Way galaxy.
- A constellation is a region of the celestial sphere with strictly defined boundaries containing a characteristic group of stars; the International Astronomical Union (1922) fixed 88 constellations.
- Apparent magnitude m characterizes a star's apparent brightness (for the Sun, m = −26.6); absolute magnitude M is the magnitude a star would have at a distance of 32.6 light-years from Earth; luminosity is the total energy a star radiates per unit time.

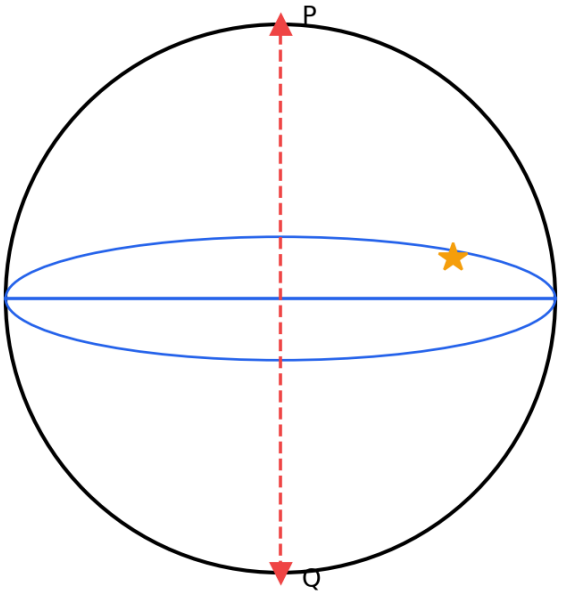
DEFINITIONS

Term	Definition
Light-year	the distance light travels in vacuum in one Earth year (≈ 9.46×10^{15} m).
Constellation	a region of the celestial sphere with strictly defined boundaries containing a characteristic group of stars.
Luminosity	the total energy radiated by a star per unit time.

FORMULAE AND RELATIONS

Formula	Note
$L = 4\pi R^2 \cdot \sigma T^4$	a star's luminosity via its radius and surface temperature (σ is the Stefan–Boltzmann constant).
1 magnitude ≈ a factor of 2.5 in brightness	a difference of 5 magnitudes corresponds to a 100-fold difference in brightness.

DIAGRAM §8



KEY POINT
Apparent magnitude does not show a star's true brightness since stars lie at different distances from Earth; absolute magnitude is used to compare true brightness.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 8, item 2

How many times brighter are second-magnitude stars than fourth-magnitude stars?

Given	SI	Solution
Δm = 4 − 2 = 2 magnitudes	*1 magnitude ≈ factor of 2.5*	brightness ratio = 2.5^Δm = 2.5^2 ≈ 6.3

Answer: About 6.3 times.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 9. The celestial sphere, celestial coordinate systems

- The celestial sphere is an imaginary sphere of arbitrary radius onto which all visible celestial bodies are projected; the world axis is the imaginary rotation axis of the celestial sphere, and its intersections with the sphere are the celestial poles.
- The equatorial coordinate system uses declination δ (analogous to latitude) and right ascension α (analogous to longitude), measured from the vernal equinox point in the constellation Aries.
- The celestial equator plane is perpendicular to the world axis and divides the celestial sphere into Northern and Southern hemispheres.

DEFINITIONS

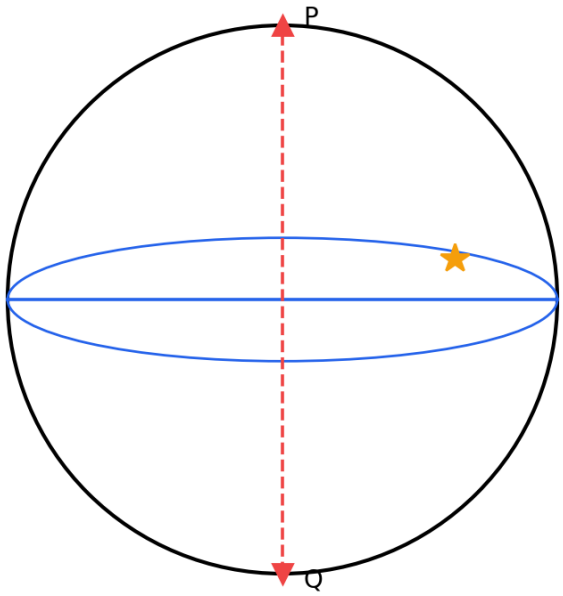
Term	Definition
Celestial sphere	an imaginary sphere of arbitrary radius onto which all visible celestial bodies are projected.

Term	Definition
Declination δ	an equatorial coordinate analogous to geographic latitude.
Right ascension α	an equatorial coordinate analogous to geographic longitude, measured from the vernal equinox point.

FORMULAE AND RELATIONS

Formula	Note
h = φ (for Polaris)	the altitude of the celestial pole above the horizon equals the geographic latitude of the site.

DIAGRAM §9



KEY POINT
A star's equatorial coordinates (unlike horizontal ones) stay essentially unchanged over 24 hours, which makes them convenient for star charts.

WORKED TEXTBOOK PROBLEM — Exercise from the section

Polaris is observed at an altitude of 51° above the horizon. Find the geographic latitude of the observation site.

Given	SI	Solution
h = 51°	$h = \varphi$	The celestial pole altitude equals the site's latitude: φ = h φ = 51°

Answer: The latitude of the site is 51°.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 10. Apparent motion of celestial bodies at various latitudes; local, zone and universal time

- At the North Pole (φ=90°), stars trace circles parallel to the horizon with Polaris at the zenith; at the equator (φ=0°), the daily paths of stars are perpendicular to the horizon and day always equals night.
- A solar day is the time between two identical culminations of the Sun; the mean solar day of 24 hours was introduced to smooth out Earth's uneven orbital motion.
- Local time depends on longitude; zone time is defined for 24 time zones of 15° each: $T = T_0 + n$, where T_0 is universal time and n is the zone number. Kazakhstan spans the 4th and 5th time zones.

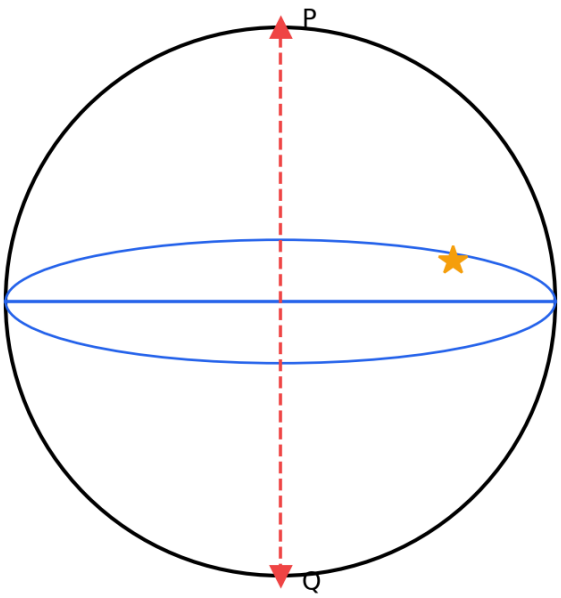
DEFINITIONS

Term	Definition
Solar day	the time interval between two upper (or lower) culminations of the Sun's centre.
Local time	the time at the same instant for points lying on the same geographic meridian.
Zone (standard) time	the mean solar time defined for 24 principal meridians spaced 15° apart in longitude.

FORMULAE AND RELATIONS

Formula	Note
24 h = 360°; 1 h = 15°; 4 min = 1°	relation between time units and angular displacement in longitude.
$T = T_0 + n$	zone time via universal time and zone number n.

DIAGRAM §10



KEY POINT
Time-zone boundaries are drawn not strictly along meridians but along state and administrative borders, for the practical convenience of timekeeping.

WORKED TEXTBOOK PROBLEM — Exercise from the section

Find the local time at a point with geographic longitude 72° East, if universal time is 14:00.

Given	SI	Solution
$T_0 = 14$ h $\lambda = 72°$	*4 min = 1°, so λ in time: 72·4 min = 288 min = 4 h 48 min*	$T = T_0 + \lambda$(time) T = 14 h + 4 h 48 min

Answer: Local time is 18:48.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 11. The laws of planetary motion in the Solar System

- Kepler's first law: every planet moves in an ellipse with the Sun at one focus.
- Kepler's second law: a planet's radius vector sweeps out equal areas in equal times (the planet moves faster near the Sun, at perihelion, and slower far from it, at aphelion).
- Kepler's third law: the squares of the orbital periods of planets are proportional to the cubes of the semi-major axes of their orbits: $T_1^2/T_2^2 = a_1^3/a_2^3$.

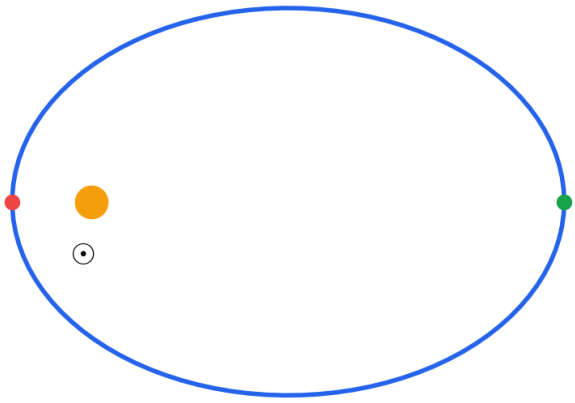
DEFINITIONS

Term	Definition
Perihelion	the point of an orbit closest to the Sun.
Aphelion	the point of an orbit farthest from the Sun.
Semi-major axis	half the longest diameter of an elliptical orbit; it determines the planet's mean distance from the Sun.

FORMULAE AND RELATIONS

Formula	Note
$\mathbf{T_1^2/T_2^2 = a_1^3/a_2^3}$	Kepler's third law.

DIAGRAM §11



KEY POINT

Kepler's third law lets you find a planet's period or distance from the Sun using the known values for another planet (e.g. for Earth: T = 1 year, a = 1 AU).

WORKED TEXTBOOK PROBLEM — Exercise from the section (using the textbook's planet table)

Mars's mean distance from the Sun is 228 million km and its period is 1.88 years. Using Kepler's third law with Earth's data (a=150 million km, T=1 year), check the consistency of these figures.

Given	SI	Solution
a_{mars} = 228 million km a_{earth} = 150 million km T_{earth} = 1 year	$T^2 \propto a^3$	$T_{mars} = T_{earth} \cdot (a_{mars}/a_{earth})^{1.5}$ $T_{mars} = 1 \cdot (228/150)^{1.5} \approx 1.87$ years

Answer: The calculated period of Mars is ≈ 1.87 years, close to the tabulated 1.88 years.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 12. Determining distances in astronomy using the parallax method

- Parallax is the angle at which a segment joining two observation points (e.g. Earth's orbital radius for annual parallax) would be seen from the star.
- The parallax method: observing an object from two points with a known baseline, the shift angle (parallax) is measured and used to calculate the distance to the object.
- A star's annual parallax is the angle at which the semi-major axis of Earth's orbit would be seen from the star; the farther the star, the smaller the parallax.

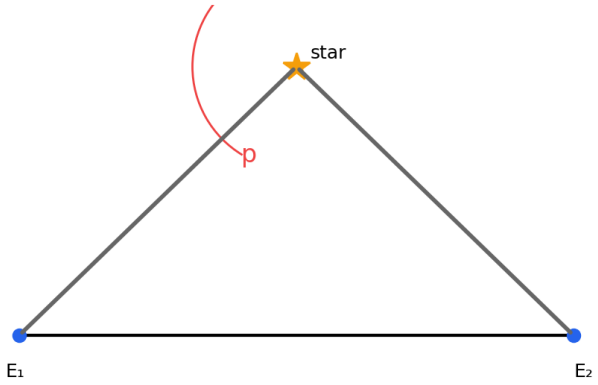
DEFINITIONS

Term	Definition
Parallax	the angle at which a segment connecting two observation points is seen from the celestial body.
Annual parallax	the angle at which the semi-major axis of Earth's orbit (mean Earth–Sun distance) is seen from the star.

FORMULAE AND RELATIONS

Formula	Note
D = a/tan p ≈ a/p (for small p in radians)	distance to the object via baseline a and parallax p.

DIAGRAM §12



KEY POINT
The parallax method only works for relatively nearby stars, since for very distant objects the parallax angle becomes too small to measure.

WORKED TEXTBOOK PROBLEM — Exercise from the section

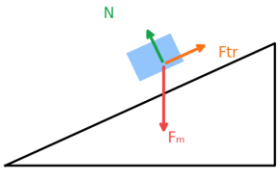
A star's annual parallax is 0.3" (arcseconds). Find the distance to the star in parsecs and approximately in light-years.

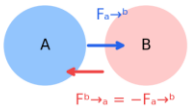
Given	SI	Solution
p = 0.3"	*D(pc) = 1/p(") — definition of the parsec*	D = 1/p = 1/0.3 ≈ 3.33 pc 1 pc ≈ 3.26 ly D ≈ 3.33·3.26 ≈ 10.9 ly

Answer: D ≈ 3.33 pc ≈ 10.9 light-years.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 3. Fundamentals of dynamics





§ 13. Newton's first law, inertial reference frames

- Newton's first law: a body retains its state of rest or uniform straight-line motion unless other bodies act on it, or their actions cancel out.
- The tendency of a body to keep its velocity when no external forces act is called inertia; a frame in which Newton's first law holds is called an inertial frame.
- Any frame moving uniformly in a straight line relative to an inertial frame is also inertial.

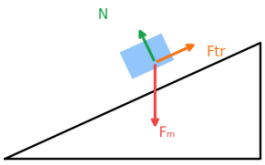
DEFINITIONS

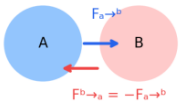
Term	Definition
Inertia	the tendency of a body to keep its velocity when no other bodies act on it, or their actions cancel.
Inertial reference frame	a frame of reference in which Newton's first law holds.

FORMULAE AND RELATIONS

Formula	Note
υ = const, if $\Sigma F = 0$	statement of Newton's first law.

DIAGRAM §13





KEY POINT
A frame attached to the Earth is approximately treated as inertial for most practical problems, though strictly it is not, due to Earth's rotation.

WORKED TEXTBOOK PROBLEM — Conceptual question with justification

A passenger standing in a uniformly moving bus leans forward when the bus brakes sharply. Explain this using Newton's first law.

Given	SI	Solution
Situation = the bus brakes, the passenger is not braced	*Qualitative explanation*	By inertia, the passenger's body tends to keep its previous speed of motion The passenger's feet are braked together with the bus floor (via friction), while the upper body keeps moving by inertia As a result the passenger leans in the direction of the previous motion

Answer: The forward lean is explained by inertia: the body keeps the velocity it had before braking.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 14. Forces in mechanics

- In mechanics we consider gravity, elastic force, and friction; all arise from the interaction of bodies.
- Elastic force arises when a body is deformed and obeys Hooke's law: F_el = kx, where k is the stiffness and x is the extension.
- Static and kinetic friction are proportional to the normal reaction force: F_fr = μN, where μ is the coefficient of friction.

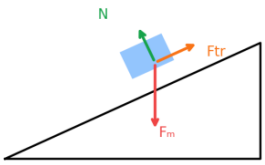
DEFINITIONS

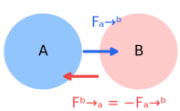
Term	Definition
Elastic force	a force that arises when a body is deformed and tends to restore its original shape.
Stiffness k	the proportionality coefficient in Hooke's law characterizing a body's elastic properties.
Coefficient of friction μ	a quantity characterizing the surfaces in contact that determines the friction force.

FORMULAE AND RELATIONS

Formula	Note
F_el = kx	Hooke's law.
F_fr = μN	friction force via the coefficient of friction and the normal reaction.

DIAGRAM §14





KEY POINT
The resultant force is the vector sum of all forces acting on a body; for uniform motion or rest, the resultant of all forces is zero.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 14, item 1

A load of mass m = 1500 kg is suspended from a cable. Find the magnitude of the elastic force in the cable.

Given	SI	Solution
m = 1500 kg	*Body at rest: F_el = mg*	F_el = mg F_el = 1500·9.8

Answer: F_el = 14700 N ≈ 14.7 kN.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 15. Newton's second law, mass

- Newton's second law: the acceleration of a body is directly proportional to the resultant applied force and inversely proportional to its mass: a = F/m.
- Mass is a measure of a body's inertia — its resistance to a change in velocity under an applied force.
- The fundamental equation of dynamics ma = F is written in projections onto chosen coordinate axes to solve specific problems.

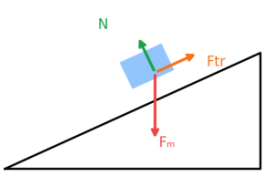
DEFINITIONS

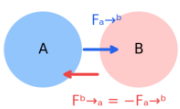
Term	Definition
Mass	a physical quantity that is a measure of a body's inertia.
Resultant force	a single force having the same effect on a body as all the applied forces combined.

FORMULAE AND RELATIONS

Formula	Note
a = F/m or F = ma	Newton's second law.
a = g(sinα − μcosα)	acceleration of a block sliding down a rough incline.

DIAGRAM §15





KEY POINT

In dynamics problems it helps to align one axis with the body's acceleration — then the equation along that axis contains ma itself, while along the perpendicular axis the sum of force projections is zero.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

With what acceleration does a block slide down an incline of angle $\alpha = 30°$ with coefficient of friction $\mu = 0.2$?

Given	SI	Solution
$\alpha = 30°$ $\mu = 0.2$	$g = 9.8\ m/s^2$	ma = −F_fr + mg·sinα (along the incline); N = mg·cosα a = g(sinα − μcosα) a = 9.8(sin30° − 0.2·cos30°) ≈ 3.3 m/s²

Answer: a ≈ 3.3 m/s².

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 16. Newton's third law

- Newton's third law: bodies act on each other with forces equal in magnitude, opposite in direction, along the same line: $F_{a\to b} = -F_{b\to a}$.
- The interaction forces in the third law act on different bodies, so they do not balance each other and never combine into a resultant.
- The third law holds both for bodies in contact and for bodies interacting at a distance (e.g. gravitationally).

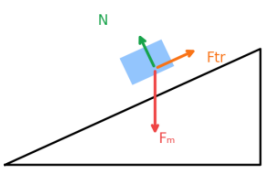
DEFINITIONS

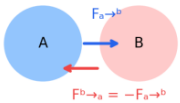
Term	Definition
Newton's third law	the forces with which two bodies act on each other are equal in magnitude, opposite in direction, and lie along the same line.

FORMULAE AND RELATIONS

Formula	Note
$F_{a\to b} = -F_{b\to a}$	mathematical form of Newton's third law.

DIAGRAM §16





KEY POINT

Action and reaction forces act on different bodies, so they never cancel within one system — otherwise no motion (like walking or a rocket launching) would be possible.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 16, item 2

A crate can be moved by applying a force of at least 60 N. Find the force the crate exerts on the floor if the friction coefficient between crate and floor is 0.3.

Given	SI	Solution
$F_{min} = 60$ N $\mu = 0.3$	*At the point of moving,* $F = F_{fr} = \mu N$	$N = F/\mu$ $N = 60/0.3 = 200$ N By Newton's third law, the crate presses on the floor with that same force N

Answer: The crate presses on the floor with a force of 200 N.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 17. The law of universal gravitation

- The law of universal gravitation: all bodies attract each other with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
- The gravitational constant $G = 6.67\times10^{-11}$ N·m²/kg² gives the force of attraction between two 1 kg bodies 1 m apart.
- The acceleration of free fall on a planet's surface is determined by its mass and radius: $g = GM/R^2$.

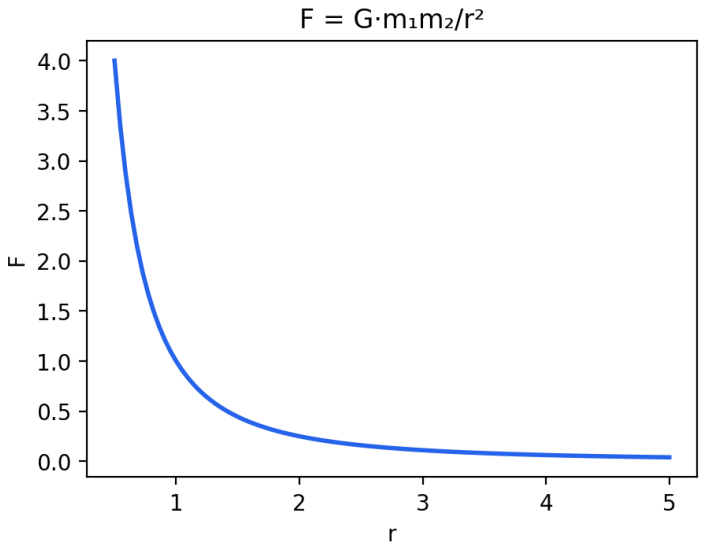
DEFINITIONS

Term	Definition
Gravitational constant G	the proportionality constant in the law of universal gravitation, $G = 6.67\times10^{-11}$ N·m²/kg².

FORMULAE AND RELATIONS

Formula	Note
$\mathbf{F = Gm_1m_2/r^2}$	the law of universal gravitation.
$\mathbf{g = GM/R^2}$	acceleration of free fall on the surface of a body of mass M and radius R.

DIAGRAM §17



KEY POINT
The distance r in the gravitation law is measured between the centres of the bodies, not their surfaces; far from a planet ($r \gg R$) the attractive force falls off quickly as $1/r^2$.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 17, item 2

Find the force acting on a 1 kg body located 600 km above Earth's surface. Earth's radius is 6400 km, Earth's mass is 6×10^{24} kg.

Given	SI	Solution
m = 1 kg h = 600 km R_E = 6400 km M = 6×10^{24} kg G = 6.67×10^{-11} N·m²/kg²	$r = R_E+h = 7\times10^6$ m	F = GMm/r² F = $6.67\times10^{-11}\cdot6\times10^{24}\cdot1/(7\times10^6)^2$ F ≈ 8.2 N

Answer: F ≈ 8.2 N.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 18. Weight of a body, weightlessness

- The weight of a body is the force with which, due to gravitational attraction, it presses on its support or suspension; unlike gravity, weight is applied to the support, not the body.
- If the support (e.g. an elevator) accelerates, weight differs from gravity: P = m(g ± a); in free fall of the support (a = g, downward), weightlessness occurs (P = 0).
- On the rotating Earth, weight at the equator is less than at the poles because of the centripetal acceleration needed to rotate with the Earth.

DEFINITIONS

Term	Definition
Weight	the force with which a body, due to gravitational attraction, acts on its support or suspension.
Weightlessness	a state in which a body's weight is zero (the support moves with the acceleration of free fall).

FORMULAE AND RELATIONS

Formula	Note
P = m(g + a)	weight in a lift accelerating upward.
P = m(g − a)	weight in a lift accelerating downward (a<g).
$a_{eq} = 4\pi^2 R_E/T^2$	centripetal acceleration of a point on Earth's equator.

DIAGRAM §18



KEY POINT
Weightlessness occurs not because gravity is absent (it still acts in orbit) but because the body and its support fall with the same acceleration and don't press on each other.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

Find the weight of a 1000-tonne body at the pole and at the equator. Take Earth's radius as 6400 km.

Given	SI	Solution
m = 1000 t = 10^6 kg R_E = 6400 km T = 24 h = 86400 s	*g ≈ 9.81 m/s² (taken the same)*	P_1 (pole) = mg = $10^6 \cdot 9.81 = 9.81 \cdot 10^6$ N $a_{eq} = 4\pi^2 R_E/T^2 \approx 0.034$ m/s² P_2 (equator) = m(g − a_{eq}) ≈ $9.77 \cdot 10^6$ N

Answer: P_1 = 9.81 MN; P_2 = 9.77 MN.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 19. Motion of a body under gravity. Motion of artificial Earth satellites

- The motion of a body thrown vertically, at an angle, or horizontally are special cases of free fall, to which the equations of uniformly variable motion with a = g apply.
- An artificial Earth satellite moves in a circular (or elliptical) orbit under gravity, which plays the role of centripetal force: $GMm/R^2 = m\upsilon^2/R$.
- The first cosmic velocity is the minimum speed at which a body becomes a satellite in a low circular orbit: $\upsilon 1 = \sqrt{(gR)} \approx 7.9$ km/s.

DEFINITIONS

Term	Definition
First cosmic velocity	the minimum speed at which a body orbits in a circle near a planet's surface without falling onto it.

FORMULAE AND RELATIONS

Formula	Note
$GMm/R^2 = m\upsilon^2/R$	condition for circular orbital motion (gravity = centripetal force).
$\upsilon 1 = \sqrt{(gR)} \approx 7.9$ km/s	the first cosmic velocity.

DIAGRAM §19



KEY POINT
An orbiting satellite is effectively in continuous free fall: it constantly "falls" toward Earth but, due to its large horizontal speed, keeps "missing" it.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 19, item 1

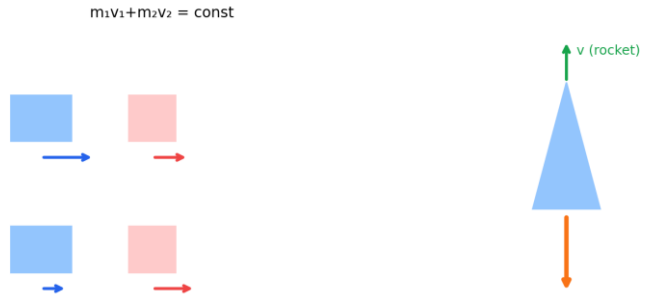
A freely falling body at a height of 100 m had a speed of 50 m/s. At what height will it be after 1 s? Take g = 10 m/s².

Given	SI	Solution
$h_0 = 100$ m $\upsilon_0 = 50$ m/s $t = 1$ s	*The body moves down; take the 0y axis downward*	$h(t) = h_0 - (\upsilon 0t + gt^2/2)$ $h(1) = 100 - (50 \cdot 1 + 10 \cdot 1^2/2)$ $h(1) = 100 - 55 = 45$ m

Answer: After 1 s the height will be 45 m.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 4. Conservation laws



§ 20. Momentum of a body and impulse of force. Conservation of momentum

- A body's momentum is a vector quantity equal to the product of mass and velocity: $p = m\upsilon$; it points in the same direction as the velocity.
- The impulse of a force Ft equals the change in momentum: $Ft = \Delta p = m\upsilon - m\upsilon_0$ — this is Newton's second law in impulse form.
- Conservation of momentum: in a closed system of bodies (no external forces, or they cancel out), the total momentum of the system stays constant.

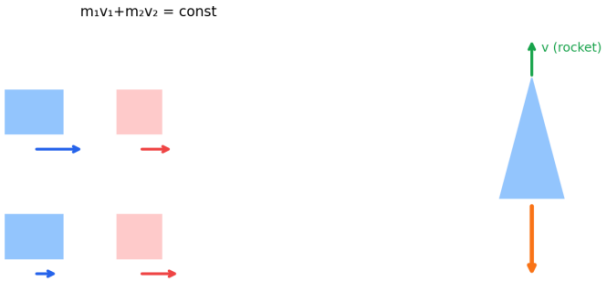
DEFINITIONS

Term	Definition
Momentum	a vector quantity equal to the product of a body's mass and velocity.
Impulse of a force	the product of a force and the time interval it acts; equal to the change in momentum.
Closed (isolated) system	a system of bodies interacting only with each other (no external forces, or they cancel).

FORMULAE AND RELATIONS

Formula	Note
$p = m\upsilon$	definition of momentum.
$Ft = \Delta p$	Newton's second law in impulse form.
$m_1\upsilon_1 + m_2\upsilon_2 = const$	conservation of momentum for a closed two-body system.

DIAGRAM §20



KEY POINT

Momentum conservation is written in projections onto a chosen axis, with signs; it holds even for collisions with energy loss (inelastic collisions), since momentum is conserved regardless of the interaction type.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

A hunter fires from a light inflatable boat. What speed does the boat gain at the moment of firing if the hunter+boat mass is 70 kg, the shot's mass is 35 g, its speed is 350 m/s, and the barrel makes a 60° angle with the horizontal?

Given	SI	Solution
$m_1 = 70$ kg $m_2 = 35$ g $\upsilon_2 = 350$ m/s $\alpha = 60°$	$m_2 = 3.5\times10^{-2}$ kg	$0 = -m_1\upsilon_1 + m_2\upsilon_2\cos\alpha$ (projection onto the firing axis) $\upsilon_1 = m_2\upsilon_2\cos\alpha/m_1$ $\upsilon_1 = 3.5\times10^{-2}\cdot350\cdot0.5/70 \approx 0.08$ m/s

Answer: $\upsilon_1 \approx 0.08$ m/s.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 21. Reactive (jet/rocket) motion

- Reactive (jet) motion is the motion that arises when part of a body's mass separates at some velocity (e.g. gas ejected from a rocket nozzle); it is explained by conservation of momentum.
- Before launch, the momentum of the "rocket + fuel" system is zero; as gas is ejected backward, the rocket gains an equal momentum directed forward.
- Reactive motion underlies rockets and jet engines, and also occurs in nature (e.g. in octopuses and squid).

DEFINITIONS

Term	Definition
Reactive motion	the motion that results when part of a body's mass separates from it at some velocity.

FORMULAE AND RELATIONS

Formula	Note
0 = m_rυ_r + m_gυ_g	conservation of momentum for a "rocket + ejected mass" system starting from rest.

DIAGRAM §21



KEY POINT

Reactive motion is possible even in a vacuum, since it needs no medium to push against — the motion arises purely from the system's own momentum conservation.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 21, item 2

At what speed did the gas escape when launching a 200 g rocket model, if its maximum height was 12.8 m? 0.5 kg of fuel was used; neglect air resistance.

Given	SI	Solution
m_model = 200 g h = 12.8 m m_fuel = 0.5 kg	*The model's launch speed from h=υ²/2g*	υ_model = √(2gh) = √(2·9.8·12.8) ≈ 15.9 m/s Conservation of momentum: m_gas·υ_gas = m_model·υ_model υ_gas = 0.2·15.9/0.5 ≈ 6.4 m/s

Answer: The gas escape speed is about 6.4 m/s.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 22. Mechanical work and energy

- The mechanical work of a force equals the product of the force magnitude, the displacement, and the cosine of the angle between them: A = Fs·cosα.
- Kinetic energy is the energy of motion: E_k = mυ²/2; the work of the resultant force equals the change in kinetic energy (the work–energy theorem).
- Potential energy is the energy of interaction between bodies: for a raised body E_p = mgh, for a deformed spring E_p = kx²/2.

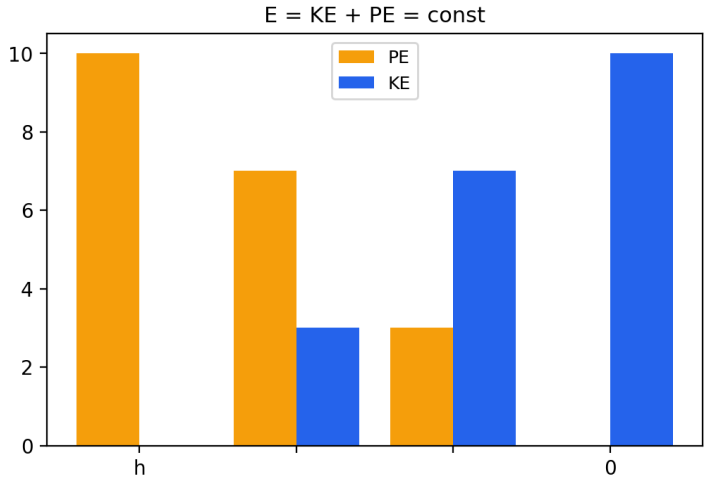
DEFINITIONS

Term	Definition
Mechanical work	the product of a force's magnitude, the body's displacement, and the cosine of the angle between them.
Kinetic energy	the energy of a moving body, equal to half the product of mass and the square of velocity.
Potential energy	the energy of interaction between bodies or parts of a body, depending on their relative positions.

FORMULAE AND RELATIONS

Formula	Note
A = Fs·cosα	mechanical work.
E_k = mυ²/2	kinetic energy.
E_p = mgh; E_p = kx²/2	potential energy of a raised body and a deformed spring.

DIAGRAM §22



KEY POINT
The work of the elastic force when stretching a spring from x_1 to x_2 is $A = -k(x_2^2 - x_1^2)/2$ (the work against it is positive, $k(x_2^2 - x_1^2)/2$), since the elastic force varies and grows with extension.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

When a 2 kg load is hung from a spring, it stretches by 4 cm. What work is needed to stretch the spring from 2 cm to 12 cm?

Given	SI	Solution
m = 2 kg x_0 = 4 cm x_1 = 2 cm x_2 = 12 cm	*$k = mg/x_0 = 2 \cdot 9.8/0.04 = 490$ N/m*	$A = k(x_2^2 - x_1^2)/2$ $A = 490 \cdot (0.12^2 - 0.02^2)/2$ $A \approx 3.5$ J

Answer: A ≈ 3.5 J.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 23. Conservation and transformation of energy

- The total mechanical energy of a body is the sum of kinetic and potential energy: E = E_k + E_p.
- Conservation of mechanical energy: in a system where only gravity and elastic forces act (no friction), total mechanical energy stays constant.
- When friction is present, part of the mechanical energy converts to internal (thermal) energy; the general law of energy conservation still holds, but mechanical energy alone is not conserved.

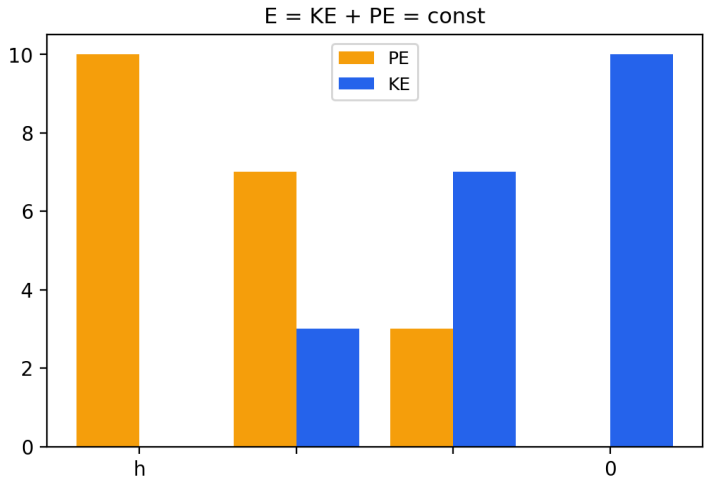
DEFINITIONS

Term	Definition
Total mechanical energy	the sum of the kinetic and potential energy of a body or system.
Conservation of mechanical energy	in a system where only gravity and elastic forces act, total mechanical energy remains unchanged.

FORMULAE AND RELATIONS

Formula	Note
E = E_k + E_p = const	conservation of total mechanical energy (no friction).
mgh = mυ²/2	potential energy converting to kinetic energy sliding from height h without friction.

DIAGRAM §23



KEY POINT
For energy-conservation problems it helps to compare the system's total energy in two states (e.g. start and end of motion), choosing the reference level for potential energy to simplify the calculation.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

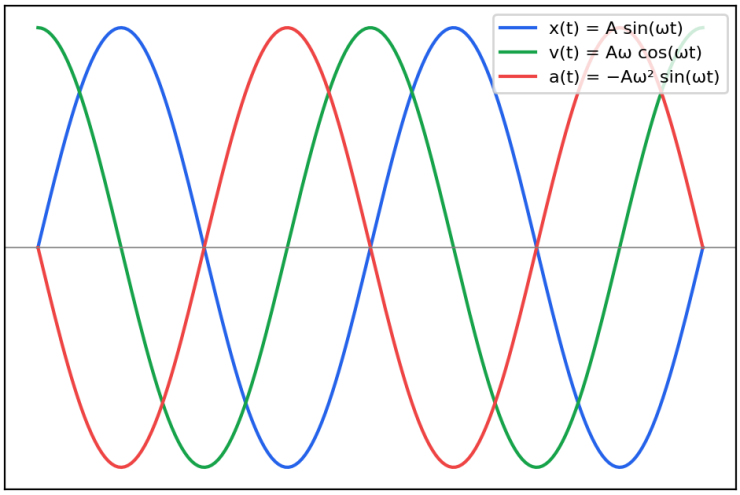
Find the speed of a body sliding down a smooth incline of height 0.8 m.

Given	SI	Solution
h = 0.8 m υ_0 = 0	*Conservation of energy: mgh = mυ²/2*	υ = √(2gh) υ = √(2·9.8·0.8) υ = 4 m/s

Answer: υ = 4 m/s.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 5. Oscillations and waves



§ 24. Oscillatory motion

- Oscillations are motions that repeat exactly or approximately at equal time intervals; free oscillations occur under the system's own internal forces after an initial push.
- Period T is the time for one complete oscillation; frequency ν = 1/T is the number of oscillations per unit time; amplitude A is the maximum displacement from equilibrium.
- Conditions for free oscillations: a restoring force directed toward equilibrium, and small damping (weak resistance).

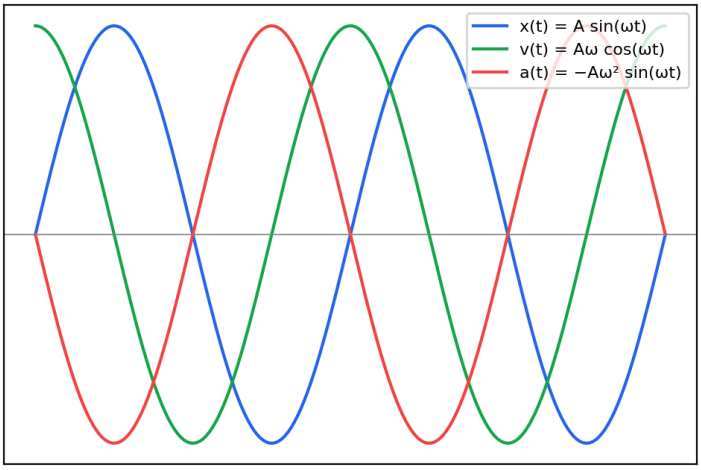
DEFINITIONS

Term	Definition
Oscillations	motions that repeat exactly or approximately at equal time intervals.
Amplitude	the maximum displacement of an oscillating body from its equilibrium position.
Free oscillations	oscillations occurring under a system's own internal forces after it is displaced from equilibrium.

FORMULAE AND RELATIONS

Formula	Note
T = t/N	period via the number of oscillations N in time t.
ν = 1/T = N/t	oscillation frequency.

DIAGRAM §24



KEY POINT
Don't confuse amplitude (the maximum displacement) with the swing of the oscillation (twice the amplitude, the distance between extreme positions).

WORKED TEXTBOOK PROBLEM — Textbook Exercise 24, item 1

A pendulum completed 50 oscillations in 1 min 40 s. Find the period and frequency.

Given	SI	Solution
N = 50 t = 1 min 40 s	*t = 100 s*	T = t/N = 100/50 = 2 s ν = 1/T = 0.5 Hz

Answer: T = 2 s; ν = 0.5 Hz.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 25. Energy transformation during oscillations. The equation of oscillatory motion

- In simple harmonic motion, kinetic and potential energy periodically convert into each other, while their sum (total mechanical energy) stays constant.
- Harmonic oscillations are described by x = A·sin(ωt) or x = A·cos(ωt), where ω = 2π/T is the angular (cyclic) frequency.
- Velocity and acceleration in SHM also vary harmonically: υ = Aω·cos(ωt), a = −Aω²·sin(ωt); the maximum values are υ_max = Aω, a_max = Aω².

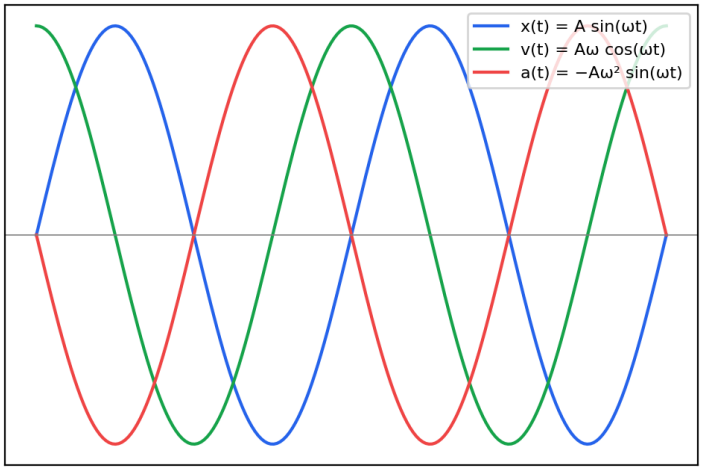
DEFINITIONS

Term	Definition
Simple harmonic motion	oscillations in which the displacement varies as a sine or cosine function.
Angular (cyclic) frequency ω	the quantity equal to the number of complete oscillations per 2π seconds; ω = 2π/T = 2πν.

FORMULAE AND RELATIONS

Formula	Note
x(t) = A·sin(ωt)	the equation of harmonic motion.
υ_max = Aω; a_max = Aω²	maximum velocity and acceleration in oscillation.

DIAGRAM §25



KEY POINT
Velocity is maximal when passing through equilibrium (x=0) and zero at the extreme points; acceleration is the opposite — maximal at the extremes, zero at equilibrium.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

From the displacement-time graph, find the maximum acceleration and speed of an oscillating body, given amplitude A = 3 cm and period T = 4 s.

Given	SI	Solution
A = 3 cm = 0.03 m T = 4 s	$\omega = 2\pi/T$	$\omega = 2\pi/4 = \pi/2 \approx 1.57$ rad/s $a_{max} = A\omega^2 = 0.03\cdot(1.57)^2 \approx 0.07$ m/s² $\upsilon_{max} = A\omega = 0.03\cdot1.57 \approx 0.05$ m/s

Answer: $a_{max} \approx 0.07$ m/s²; $\upsilon_{max} \approx 0.05$ m/s.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 26. Oscillations of simple and spring pendulums

- A simple (mathematical) pendulum is a point mass on a weightless inextensible string; its period $T = 2\pi\sqrt{(l/g)}$ depends only on the string length and g.
- A spring pendulum is a mass on a spring; its period $T = 2\pi\sqrt{(m/k)}$ depends on the mass and spring stiffness, but not on amplitude.
- The independence of period from amplitude (for small oscillations) is called isochronism — this property underlies clock mechanisms.

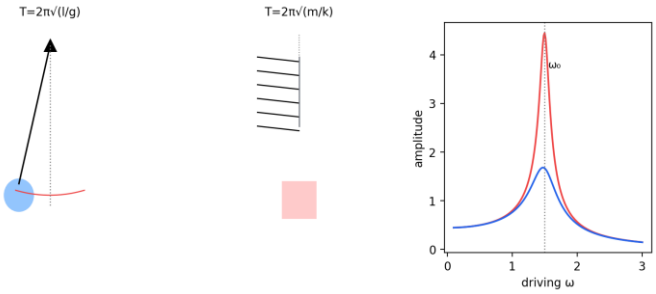
DEFINITIONS

Term	Definition
Simple (mathematical) pendulum	a point mass suspended by a weightless, inextensible string.
Spring pendulum	a mass attached to a spring, capable of oscillating.
Isochronism	the independence of the oscillation period from its amplitude.

FORMULAE AND RELATIONS

Formula	Note
T = 2π√(l/g)	period of a simple pendulum.
T = 2π√(m/k)	period of a spring pendulum.

DIAGRAM §26



KEY POINT
The simple pendulum's period formula only holds for small angles of deflection (up to about 5°); at larger amplitudes the period starts depending on amplitude.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

Find the length of a pendulum whose period is 2 s.

Given	SI	Solution
T = 2 s g = 9.8 m/s²	$T^2 = 4\pi^2 l/g$	$l = T^2 g/(4\pi^2)$ $l = 4 \cdot 9.8/(4 \cdot 3.14^2)$ $l \approx 1$ m

Answer: l ≈ 1 m.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 27. Free and forced oscillations, resonance

- The free oscillations of a real pendulum gradually damp out due to friction and resistance — energy is lost and amplitude decreases.
- Forced oscillations occur under a periodically varying external force; once established, they occur at the frequency of that force, not the system's natural frequency.
- Resonance is a sharp increase in the amplitude of forced oscillations when the driving frequency matches (or is close to) the system's natural frequency.

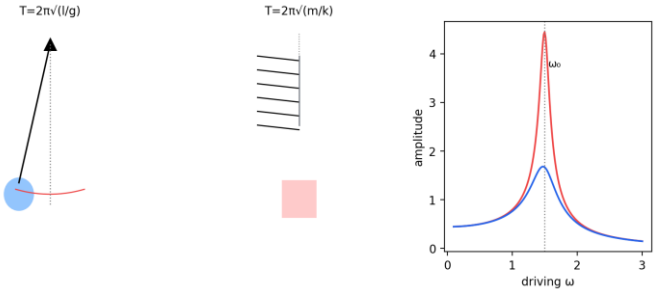
DEFINITIONS

Term	Definition
Damped oscillations	oscillations whose amplitude gradually decreases due to energy losses from friction and resistance.
Forced oscillations	oscillations occurring under an external periodically varying force.
Resonance	a sharp rise in the amplitude of forced oscillations when the driving frequency matches the system's natural frequency.

FORMULAE AND RELATIONS

Formula	Note
ω_driving = ω₀ (resonance condition)	resonance occurs when the driving frequency equals the system's natural frequency.

DIAGRAM §27



KEY POINT
Resonance can be useful (tuning a radio receiver, musical instruments) or destructive (bridges collapsing from synchronized footsteps or wind) — so engineers often avoid a structure's resonant frequencies.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 27, item 1

As a train moves on 25 m rails, the car gets jolted at the joints, causing forced oscillations. At what train speed does resonance occur, if the car's natural vertical oscillation period is 1.25 s?

Given	SI	Solution
l = 25 m T = 1.25 s	*Resonance: the time between jolts equals T*	υ = l/T υ = 25/1.25 υ = 20 m/s

Answer: υ = 20 m/s (72 km/h).

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 28. Free electromagnetic oscillations

- An oscillating (LC) circuit consists of a capacitor of capacitance C and a coil of inductance L; free electromagnetic oscillations of charge, current, and voltage occur in it.
- During oscillation, energy periodically transfers between the capacitor's electric field and the coil's magnetic field, analogous to potential-to-kinetic energy conversion in a spring pendulum.
- The period of free oscillations in the circuit is given by Thomson's formula: T = 2π√(LC).

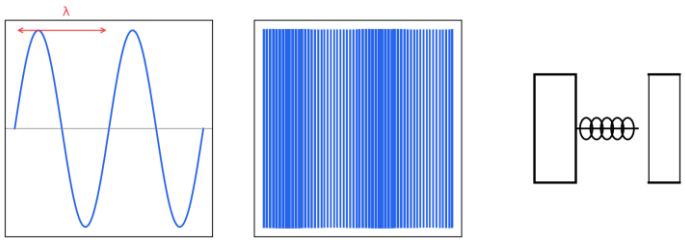
DEFINITIONS

Term	Definition
Oscillating (LC) circuit	an electric circuit consisting of a capacitor and an inductor, in which electromagnetic oscillations can occur.
Inductance L	a physical quantity characterizing a coil's ability to create magnetic flux and oppose changes in current.

FORMULAE AND RELATIONS

Formula	Note
$T = 2\pi\sqrt{(LC)}$	Thomson's formula for the period of oscillation in the circuit.
$W_{el} = q^2/(2C)$; $W_{mag} = LI^2/2$	energy of the capacitor's electric field and the coil's magnetic field.

DIAGRAM §28



KEY POINT
Analogy with mechanical oscillations: charge q corresponds to displacement x, current I to velocity υ, inductance L to mass m, and 1/C to spring stiffness k.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 28, item 2

What is the period of natural oscillations in a circuit with inductance 2.5 mH and capacitance 1.5 μF?

Given	SI	Solution
L = 2.5 mH C = 1.5 μF	*$L = 2.5\times10^{-3}$ H; $C = 1.5\times10^{-6}$ F*	$T = 2\pi\sqrt{(LC)}$ $T = 2\cdot3.14\cdot\sqrt{(2.5\times10^{-3}\cdot1.5\times10^{-6})}$ $T \approx 1.2\times10^{-4}$ s

Answer: $T \approx 1.2\times10^{-4}$ s (0.12 ms).

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 29. Wave motion

- A mechanical wave is the process of oscillations propagating through an elastic medium; matter itself is not transported, only the energy and momentum of the oscillations.
- In a transverse wave, particles oscillate perpendicular to the direction of propagation (water surface waves, a string); in a longitudinal wave, along the direction of propagation (sound in air).
- Wavelength λ is the distance between the nearest points oscillating in phase; it relates to speed and period (frequency): $\lambda = \upsilon T = \upsilon/\nu$.

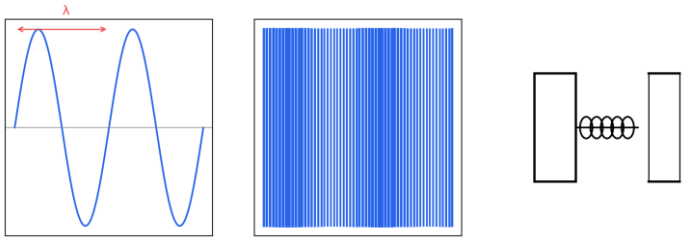
DEFINITIONS

Term	Definition
Mechanical wave	the process of oscillations propagating through an elastic medium without transport of matter.
Wavelength λ	the distance between the nearest points of a medium oscillating in the same phase.

FORMULAE AND RELATIONS

Formula	Note
$\lambda = \upsilon T = \upsilon/\nu$	relation between wavelength, speed, period and frequency.

DIAGRAM §29



KEY POINT
Wave speed is determined by the medium's properties (elasticity, density) and stays constant across different frequencies in the same medium; wavelength then varies inversely with frequency.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 29, item 1

Find the wavelength of a wave from a source oscillating at 0.165 kHz, propagating at 330 m/s.

Given	SI	Solution
ν = 0.165 kHz υ = 330 m/s	*ν = 165 Hz*	$\lambda = \upsilon/\nu$ λ = 330/165 λ = 2 m

Answer: λ = 2 m.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 30. Sound, characteristics of sound, acoustic resonance, echo

- Sound is a longitudinal mechanical wave in an elastic medium, perceived by the human ear roughly between 16 Hz and 20 kHz; sound does not propagate in a vacuum.
- Pitch is determined by frequency, loudness by amplitude, and timbre by the set of frequencies (overtones) accompanying the fundamental tone.
- An echo is a reflected sound wave perceived separately from the direct sound; acoustic resonance is a sharp amplification of sound when the driving frequency matches the resonator's natural frequency.

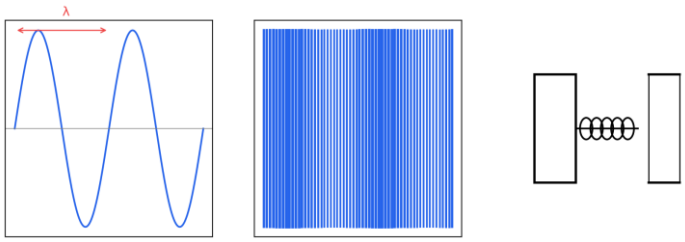
DEFINITIONS

Term	Definition
Sound	a longitudinal mechanical wave perceived by the ear in the range of roughly 16 Hz to 20 kHz.
Echo	a sound wave reflected from an obstacle and perceived separately from the direct sound.

FORMULAE AND RELATIONS

Formula	Note
s = υ_sound·t	distance to a reflecting surface via the round-trip time (t is total time; one-way distance s = υt/2).
υ_sound ≈ 340 m/s (in air)	the speed of sound in air under normal conditions.

DIAGRAM §30



KEY POINT
For an echo to be heard as a separate sound, the reflected wave must arrive at least 0.1 s after the direct sound — otherwise the ear can't distinguish the two signals.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 30, item 3

What is the sea depth if an ultrasonic signal, reflected from the seabed, returns after 1.2 s?

Given	SI	Solution
t = 1.2 s υ_sound (in water) = 1500 m/s	*The signal travels the distance twice (there and back)*	s = υ·t/2 s = 1500·1.2/2 s = 900 m

Answer: The sea depth is 900 m.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 31. Electromagnetic waves. The electromagnetic wave spectrum

- An electromagnetic wave is a propagating alternating electromagnetic field; unlike mechanical waves, it can travel through a vacuum at the speed of light c ≈ 3×10^8 m/s.
- Electromagnetic waves are transverse: the electric and magnetic field vectors oscillate perpendicular to each other and to the direction of propagation.
- The electromagnetic spectrum (in order of increasing frequency): radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, gamma rays — all of the same nature but differing in wavelength and how they are produced.

DEFINITIONS

Term	Definition
Electromagnetic wave	a propagating alternating electromagnetic field.
Speed of electromagnetic waves in vacuum	a constant c ≈ 3×10^8 m/s, independent of the wave's frequency.

FORMULAE AND RELATIONS

Formula	Note
c = λν	relation between the speed of light, wavelength and frequency in vacuum.

DIAGRAM §31



KEY POINT
All electromagnetic waves travel through vacuum at the same speed c, so higher-frequency waves have shorter wavelengths: λ = c/ν.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 31, item 1

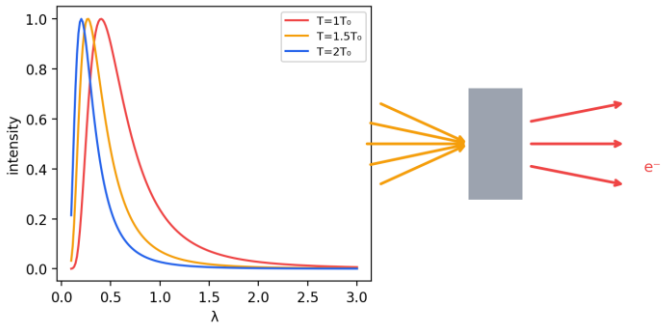
How far from a radar antenna is an object if the reflected radio signal returned after 200 µs?

Given	SI	Solution
t = 200 µs c = 3×10⁸ m/s	*t = 2×10⁻⁴ s; the signal covers the distance twice*	s = ct/2 s = 3×10⁸·2×10⁻⁴/2 s = 3×10⁴ m

Answer: s = 30 km.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 6. Atomic structure, atomic phenomena



§ 32. Thermal radiation, light as an electromagnetic wave

- All heated bodies emit electromagnetic waves due to the thermal motion of particles; the spectrum and intensity of thermal radiation depend on the body's temperature.
- Attempts to explain blackbody radiation within classical physics led to the "ultraviolet catastrophe"; Max Planck resolved this by proposing that energy is emitted in discrete portions (quanta): $E = h\nu$.
- Light exhibits both the properties of an electromagnetic wave (interference, diffraction) and a stream of particle-quanta (the photoelectric effect) — this is called wave-particle duality.

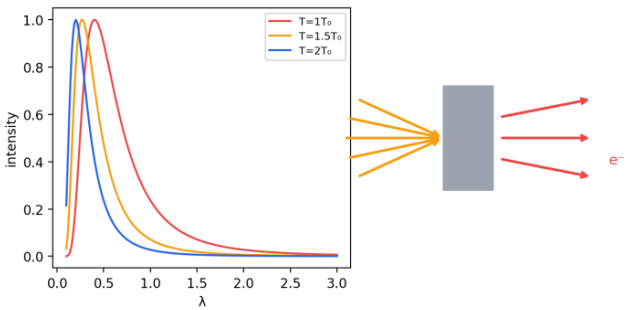
DEFINITIONS

Term	Definition
Thermal radiation	electromagnetic radiation emitted by a substance due to its internal (thermal) energy.
Quantum of energy	the smallest portion of energy exchanged between matter and the electromagnetic field; $E = h\nu$.
Planck's constant h	a fundamental physical constant, $h \approx 6.63 \times 10^{-34}$ J·s, relating a quantum's energy to the radiation frequency.

FORMULAE AND RELATIONS

Formula	Note
$E = h\nu$	the energy of a quantum (photon) via the radiation frequency.
$E = hc/\lambda$	photon energy via wavelength.

DIAGRAM §32



KEY POINT
Planck's quantum hypothesis was the starting point of quantum physics: it first showed that at the microscopic level, energy cannot be transferred continuously, only in discrete portions.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

Find the energy of a photon corresponding to a wavelength of 500 nm (green light).

Given	SI	Solution
$\lambda = 500$ nm $h = 6.63\times10^{-34}$ J·s $c = 3\times10^{8}$ m/s	$\lambda = 5\times10^{-7}$ m	$E = hc/\lambda$ $E = 6.63\times10^{-34}\cdot3\times10^{8}/(5\times10^{-7})$ $E \approx 3.98\times10^{-19}$ J ≈ 2.5 eV

Answer: $E \approx 3.98\times10^{-19}$ J (≈ 2.5 eV).

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 33. The photoelectric effect and its laws, applications

- The photoelectric effect is the ejection of electrons from a substance under light; discovered by H. Hertz, its laws established by A. Stoletov, and explained by A. Einstein using the quantum theory of light.
- Einstein's photoelectric equation: the absorbed photon's energy goes toward the work function needed to free the electron and toward its kinetic energy: hν = A_wf + E_k.
- The red (threshold) limit of the photoelectric effect ν_0 is the minimum light frequency at which the effect still occurs for a given substance ($h\nu_0$ = A_wf); below ν_0, no photoeffect occurs regardless of intensity.

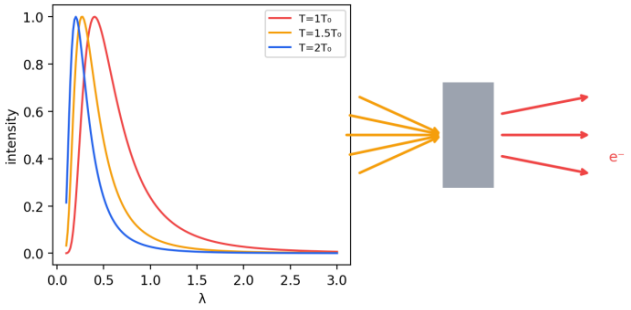
DEFINITIONS

Term	Definition
Photoelectric effect	the ejection of electrons from a substance under the action of light.
Work function A_wf	the minimum energy needed to remove an electron from a substance.
Threshold (red limit) of photoeffect	the minimum frequency (maximum wavelength) of light at which the photoeffect still occurs.

FORMULAE AND RELATIONS

Formula	Note
$h\nu = A_{wf} + E_k$	Einstein's equation for the photoelectric effect.
$\nu_0 = A_{wf}/h$	the threshold frequency of the photoelectric effect.

DIAGRAM §33



KEY POINT
The kinetic energy of ejected electrons depends only on the light's frequency, not its intensity; intensity only affects the number of ejected electrons (photocurrent) — a key quantum effect inexplicable by wave theory alone.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

The work function for zinc is 3.74 eV. Find the threshold frequency of the photoelectric effect for zinc.

Given	SI	Solution
$A_{wf} = 3.74$ eV $h = 6.63\times10^{-34}$ J·s	$A_{wf} = 3.74\cdot1.6\times10^{-19} \approx 5.98\times10^{-19}$ J	$\nu_0 = A_{wf}/h$ $\nu_0 = 5.98\times10^{-19}/6.63\times10^{-34}$ $\nu_0 \approx 9\times10^{14}$ Hz

Answer: $\nu_0 \approx 9\times10^{14}$ Hz.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 34. The structure of the atom. Rutherford's experiments

- In experiments scattering alpha particles off thin gold foil, Rutherford found that some particles deflected through large angles (up to 180°) — explainable only by a small, dense, positively charged nucleus inside the atom.
- Rutherford's planetary (nuclear) model: nearly all the atom's mass and its entire positive charge are concentrated in a tiny nucleus (radius ≈ 10^{-15} m), with electrons orbiting it; the atom's overall size is ≈ 10^{-10} m.
- The planetary model could not explain the atom's stability or the discrete (line) emission spectra of atoms from the standpoint of classical electrodynamics.

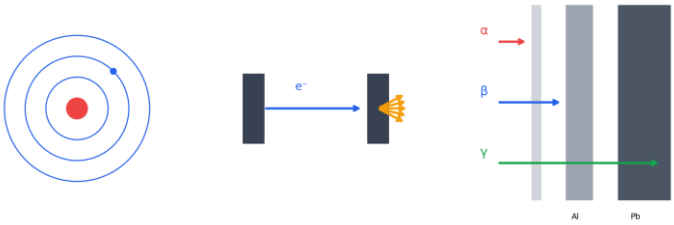
DEFINITIONS

Term	Definition
Atomic nucleus	the central part of an atom that holds nearly all its mass and its entire positive charge.
Planetary model of the atom	a model in which electrons orbit a small, massive, positively charged nucleus, like planets around the Sun.

FORMULAE AND RELATIONS

Formula	Note
$R_{nucleus} \approx 10^{-15}$ m; $R_{atom} \approx 10^{-10}$ m	characteristic sizes of the nucleus and the atom.

DIAGRAM §34



KEY POINT
The fact that most alpha particles passed through the foil with almost no deflection showed the atom is mostly empty space; only rare particles passing close to the nucleus deflected strongly.

WORKED TEXTBOOK PROBLEM — Conceptual question with justification

Why, when firing alpha particles at thin gold foil, did only a small fraction deflect through large angles while most passed through almost undeflected?

Given	SI	Solution
Observation = most particles pass straight through, rare ones deflect strongly	*Qualitative analysis*	Most of the atom's volume is empty, so most alpha particles meet no obstacle The positive charge and nearly all the atom's mass are concentrated in a very small volume — the nucleus Only particles passing close to the nucleus experience strong Coulomb repulsion and deflect through large angles

Answer: This proves the atom is mostly empty, with its mass and charge concentrated in a compact nucleus.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 35. Bohr's quantum postulates, the Bohr model of the hydrogen atom

- Bohr's first postulate (stationary states): an atom can exist only in special stationary states of definite energy, in which it does not radiate, even though the electron accelerates.
- Bohr's second postulate (frequencies): when an atom transitions between stationary states, one photon is emitted or absorbed with energy equal to the difference between the state energies: $h\nu = E_n - E_m$.
- The Bohr model of the hydrogen atom explained hydrogen's line emission spectrum and allowed calculation of allowed orbital radii and electron energy levels.

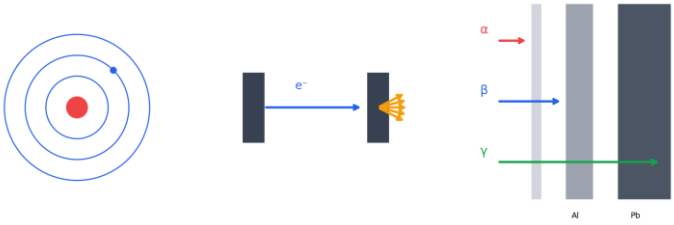
DEFINITIONS

Term	Definition
Stationary state	a special state of an atom with definite energy, in which the atom does not radiate.
Energy level	a value of energy that an atom may have in one of its stationary states.

FORMULAE AND RELATIONS

Formula	Note
$h\nu = E_n - E_m$	Bohr's second postulate (the frequency rule).
$E_n = -E_1/n^2$	the energy of the n-th stationary level of the hydrogen atom (E_1 is the ground-state energy).

DIAGRAM §35



KEY POINT
Bohr's postulates combined classical mechanics (electron orbiting) with quantum restrictions (only certain orbits are allowed) — an important, though not final, step toward a consistent quantum mechanics.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

Find the emission frequency when an electron in a hydrogen atom transitions from the third to the second energy level, given $E_2 = -3.4$ eV, $E_3 = -1.5$ eV.

Given	SI	Solution
$E_2 = -3.4$ eV $E_3 = -1.5$ eV	$\Delta E = E_3 - E_2 = 1.9$ eV = 3.04×10^{-19} J	$h\nu = E_3 - E_2$ $\nu = \Delta E/h = 3.04\times10^{-19}/6.63\times10^{-34}$ $\nu \approx 4.6\times10^{14}$ Hz

Answer: $\nu \approx 4.6\times10^{14}$ Hz.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 36. Lasers, applications of lasers

- A laser is a source of coherent, monochromatic, narrowly directed radiation, based on the stimulated (induced) emission of light by atoms.
- Laser operation requires a population inversion — a state where more atoms occupy the upper energy level than the lower one, achieved by external energy "pumping".
- Lasers are used in medicine (surgery, diagnostics), industry (cutting, welding, marking), communications (fibre-optic lines), measurement technology, and reading/writing information.

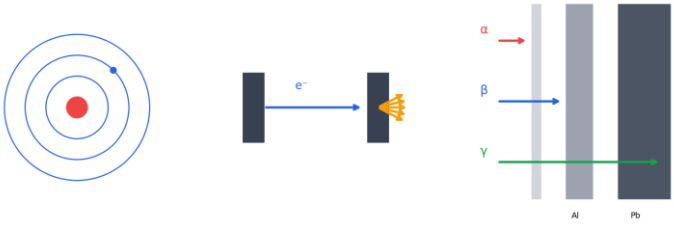
DEFINITIONS

Term	Definition
Stimulated emission	the emission of a photon by an atom triggered by an external photon of the same frequency, producing a new photon identical in direction, frequency, and phase.
Population inversion	a state of matter in which more particles occupy a higher energy level than a lower one.

FORMULAE AND RELATIONS

Formula	Note
E = hν (for each emitted photon)	the energy of photons emitted by stimulated emission is the same and equals the level energy difference.

DIAGRAM §36



KEY POINT
Laser light differs from ordinary light in its coherence (phase alignment) and high monochromaticity, which allows a narrow, non-diverging, high-power beam.

WORKED TEXTBOOK PROBLEM — Exercise from the section

A helium-neon laser emits light with wavelength 632.8 nm. Find the energy of one photon of this radiation.

Given	SI	Solution
λ = 632.8 nm h = 6.63×10^{-34} J·s c = 3×10^{8} m/s	*λ = 6.328×10^{-7} m*	E = hc/λ E = 6.63×10^{-34}·3×10^{8}/(6.328×10^{-7}) E ≈ 3.14×10^{-19} J ≈ 1.96 eV

Answer: E ≈ 3.14×10^{-19} J (≈ 1.96 eV).

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 7. The atomic nucleus



§ 37. Composition of the atomic nucleus, nuclear forces, binding energy

- The atomic nucleus consists of protons (charge +e, mass ≈ 1 amu) and neutrons (no charge, mass ≈ 1 amu), together called nucleons; the number of protons Z equals the element's atomic number, and mass number A = Z + N.
- Nuclear forces are short-range (≈ 10^{-15} m) attractive forces between nucleons, far exceeding the Coulomb repulsion between protons, and are responsible for the nucleus's stability.
- The binding energy of a nucleus is the energy required to completely separate it into individual nucleons; it relates to the mass defect Δm (the difference between the sum of nucleon masses and the nucleus's mass) via Einstein's formula $E_b = \Delta m \cdot c^2$.

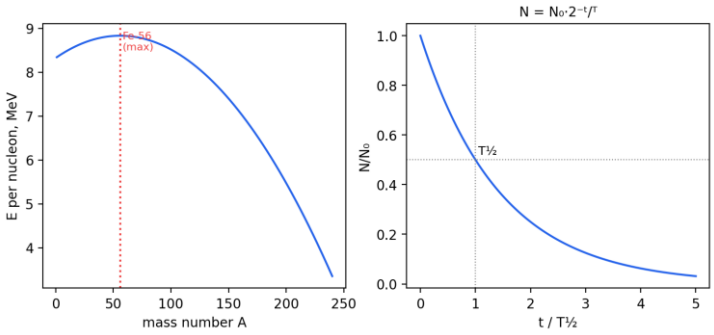
DEFINITIONS

Term	Definition
Nucleons	the collective name for protons and neutrons as constituent particles of the atomic nucleus.
Mass defect	the difference between the sum of the masses of separate nucleons and the mass of the nucleus they form.
Binding energy of a nucleus	the energy required to completely split a nucleus into its separate nucleons.

FORMULAE AND RELATIONS

Formula	Note
A = Z + N	mass number via the numbers of protons Z and neutrons N.
$E_b = \Delta m \cdot c^2$	the binding energy of a nucleus via the mass defect (Einstein's formula).
$\Delta m = Zm_p + Nm_n - m_{nucleus}$	mass defect via the proton, neutron, and nucleus masses.

DIAGRAM §37



KEY POINT
Binding energy per nucleon peaks for medium-mass nuclei (near iron, A≈56) — which is why both fission of heavy nuclei and fusion of light nuclei release energy.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

Find the binding energy of a helium nucleus ${}^{4}_{2}He$, given nucleus mass 4.00150 amu, proton mass 1.00728 amu, neutron mass 1.00867 amu (1 amu = 931.5 MeV).

Given	SI	Solution
Z = 2 N = 2 m_nucleus = 4.00150 amu m_p = 1.00728 amu m_n = 1.00867 amu	*1 amu = 931.5 MeV*	Δm = 2·1.00728 + 2·1.00867 − 4.00150 Δm = 0.03040 amu E_b = 0.03040·931.5 ≈ 28.3 MeV

Answer: E_b ≈ 28.3 MeV.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 38. Radioactivity, the law of radioactive decay

- Radioactivity is the spontaneous transformation of unstable nuclei of one element into nuclei of another element, accompanied by the emission of particles or radiation.
- The main types of radioactive decay: alpha decay (emission of a helium nucleus), beta decay (emission of an electron or positron), gamma radiation (high-energy photons, usually accompanying alpha or beta decay).
- The law of radioactive decay: the number of undecayed nuclei decreases exponentially with time, $N = N_0 \cdot 2^{-t/T}$, where T is the half-life, independent of external conditions.

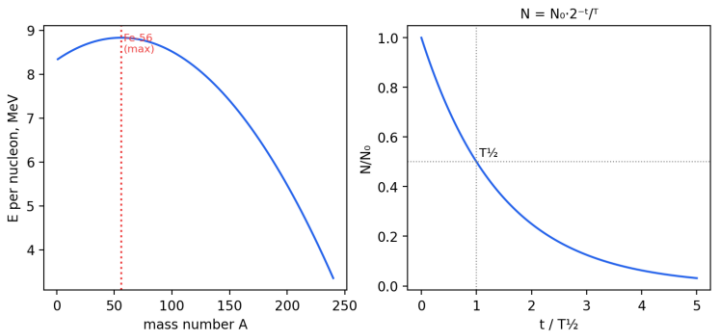
DEFINITIONS

Term	Definition
Radioactivity	the spontaneous transformation of unstable atomic nuclei into nuclei of other elements, with the emission of particles or radiation.
Half-life T	the time in which half of the original number of radioactive nuclei decays.

FORMULAE AND RELATIONS

Formula	Note
$N = N_0 \cdot 2^{-t/T}$	the law of radioactive decay.
${}^{4}_{2}{}^{R}ZX \rightarrow {}^{44-1}_{2}{}^{R}Z_{-2}Y + {}^{4}_{2}He$	alpha decay scheme: mass number decreases by 4, nuclear charge by 2.

DIAGRAM §38



KEY POINT
Half-life is a statistical property of a huge number of nuclei; the decay moment of a single nucleus cannot be predicted, but the decay law holds with high accuracy for a large number of nuclei.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 38, item 1

The half-life of a radioactive isotope is 8 days. After how long will 75% of the original number of nuclei have decayed?

Given	SI	Solution
T = 8 days fraction decayed = 75% → 25% remains	$N/N_0 = 0.25 = 1/4$	$N/N_0 = 2^{-t/T} = 1/4 = 2^{-2}$ $t/T = 2$ $t = 2T = 16$ days

Answer: After 16 days.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 39. Nuclear reactions, nuclear fission

- A nuclear reaction is the transformation of atomic nuclei through interaction with other nuclei or particles; charge, mass number, energy, and momentum are all conserved.
- Nuclear fission is the splitting of a heavy nucleus (e.g. uranium-235) into two lighter nuclei when struck by a neutron, releasing energy and several new neutrons.
- A chain fission reaction occurs when neutrons released by one nucleus's fission trigger fission in neighbouring nuclei; with a neutron multiplication factor k>1 the reaction grows explosively (the basis of nuclear weapons, and, controlled at k=1, of nuclear reactors).

DEFINITIONS

Term	Definition
Nuclear reaction	the transformation of atomic nuclei through interaction with other nuclei or elementary particles.

Term	Definition
Chain reaction	a reaction in which the particles that trigger it (e.g. neutrons) are produced as its own products.

FORMULAE AND RELATIONS

Formula	Note
$_{92}^{235}U + _{0}^{1}n \rightarrow$ fragments + $k\cdot _{0}^{1}n + E$	scheme of uranium-235 fission triggered by a neutron, releasing energy E.
conservation laws: ΣZ, ΣA	the sums of charge and mass numbers before and after the reaction are equal.

DIAGRAM §39



KEY POINT
Energy is released in fission because the fragments' total binding energy per nucleon exceeds that of the original heavy nucleus (the fragments lie closer to the peak of the binding-energy curve near iron).

WORKED TEXTBOOK PROBLEM — Textbook Exercise 39, item 1

Fill in the missing nuclide in the nuclear reaction: $_{94}^{239}Pu + _{0}^{1}n \rightarrow _{54}^{140}Xe + ? + 2_{0}^{1}n$.

Given	SI	Solution
Reagents = $_{94}^{239}Pu$ (A=239, Z=94) + $_{0}^{1}n$ Products = $_{54}^{140}Xe$ (A=140, Z=54) + ? + $2_{0}^{1}n$	*Conservation of A and Z*	A: 239+1 = 140 + A? + 2 → A? = 98 Z: 94+0 = 54 + Z? + 0 → Z? = 40 The element with Z=40 is zirconium Zr

Answer: The missing product is ^{4098}Zr (zirconium-98).

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 40. Thermonuclear reaction, nuclear power engineering

- A thermonuclear reaction is the fusion of light atomic nuclei into a heavier one at extremely high temperatures (tens of millions of kelvin), releasing a large amount of energy.
- On the Sun and stars, hydrogen fuses into helium; helium's binding energy per nucleon is greater than hydrogen's, so this fusion releases energy.
- A nuclear power plant uses a controlled chain fission reaction of uranium in a reactor: the fission energy heats a coolant, which via a steam generator drives a turbine and electric generator.

DEFINITIONS

Term	Definition
Thermonuclear (fusion) reaction	the fusion of light atomic nuclei into a heavier nucleus at extremely high temperatures.
Nuclear reactor	a device in which a controlled chain fission reaction of nuclei is carried out with energy release.

FORMULAE AND RELATIONS

Formula	Note
$_1^2H + _1^3H \rightarrow _2^4He + _0^1n + E$	the deuterium-tritium fusion reaction (an example of a thermonuclear reaction).
$E = \Delta m \cdot c^2$	the energy released in a fusion reaction via the mass defect.

DIAGRAM §40



KEY POINT
Fusion on Earth requires extremely high temperatures to overcome the Coulomb repulsion between like-charged nuclei; controlled fusion remains an active research topic rather than an industrial technology today.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

In the deuterium-tritium fusion reaction $_1^2H + _1^3H \rightarrow _2^4He + _0^1n$, 17.6 MeV of energy is released. Find the mass defect of this reaction.

Given	SI	Solution
E = 17.6 MeV	*1 amu = 931.5 MeV*	$\Delta m = E/931.5$ $\Delta m = 17.6/931.5$ $\Delta m \approx 0.0189$ amu

Answer: $\Delta m \approx 0.0189$ amu.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 41. Elementary particles

- Elementary particles are the smallest known particles of matter, believed to have no internal structure (by current understanding), or, for some, to be made of even more fundamental particles (quarks).
- Elementary particles include the photon, electron, proton, neutron (the latter two made of quarks), neutrinos, and many other particles discovered in accelerator experiments.

- Every particle has a corresponding antiparticle with the same mass but opposite charge and certain other quantum numbers; when a particle meets its antiparticle, they annihilate, releasing energy as photons.

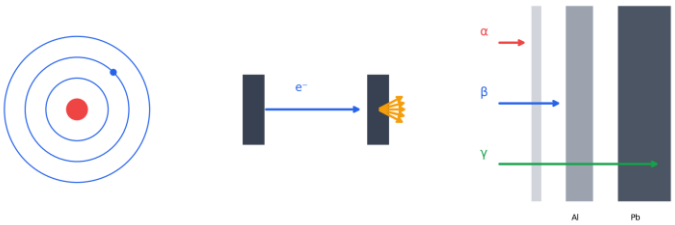
DEFINITIONS

Term	Definition
Elementary particle	the smallest known particle of matter, with no internal structure by current understanding, or built from more fundamental particles.
Antiparticle	a particle with the same mass as its corresponding particle but opposite electric charge and certain other quantum numbers.
Annihilation	the process by which a particle and its antiparticle convert into radiation (photons) upon collision.

FORMULAE AND RELATIONS

Formula	Note
$e^- + e^+ \rightarrow 2\gamma$	example: electron-positron annihilation producing two photons.
$E = 2m_ec^2$	the minimum energy released in electron-positron annihilation.

DIAGRAM §41



KEY POINT
Modern particle physics (the Standard Model) describes matter through quarks and leptons, and their interactions through carrier particles (photon, gluons, W and Z bosons); this field continues to actively develop.

WORKED TEXTBOOK PROBLEM — Worked example from the textbook

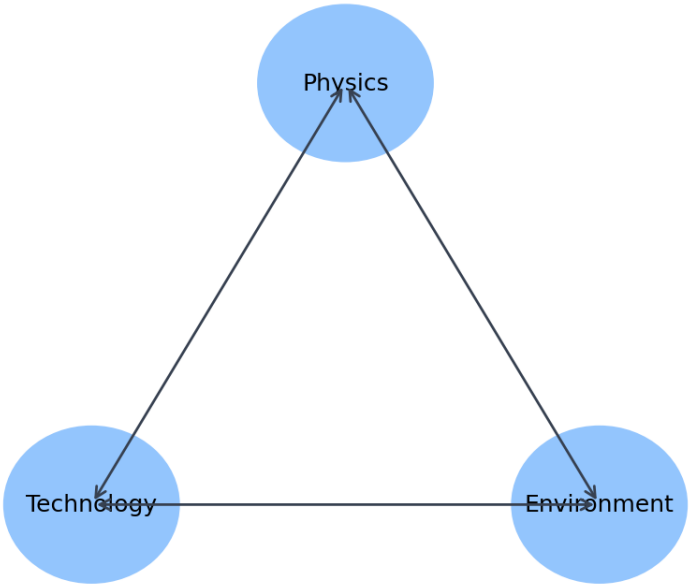
When an electron and positron at rest annihilate, two identical photons form. Find the energy and frequency of each photon ($m_e = 9.1\times10^{-31}$ kg).

Given	SI	Solution
$m_e = 9.1\times10^{-31}$ kg $c = 3\times10^8$ m/s $h = 6.63\times10^{-34}$ J·s	*The total rest energy $2m_ec^2$ splits equally between the two photons*	$E_{photon} = m_ec^2 = 9.1\times10^{-31}\cdot(3\times10^8)^2 \approx 8.19\times10^{-14}$ J $\nu = E/h = 8.19\times10^{-14}/6.63\times10^{-34}$ $\nu \approx 1.23\times10^{20}$ Hz

Answer: $E \approx 8.19\times10^{-14}$ J (0.51 MeV); $\nu \approx 1.23\times10^{20}$ Hz.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 8. Conclusion. The modern physical picture of the world



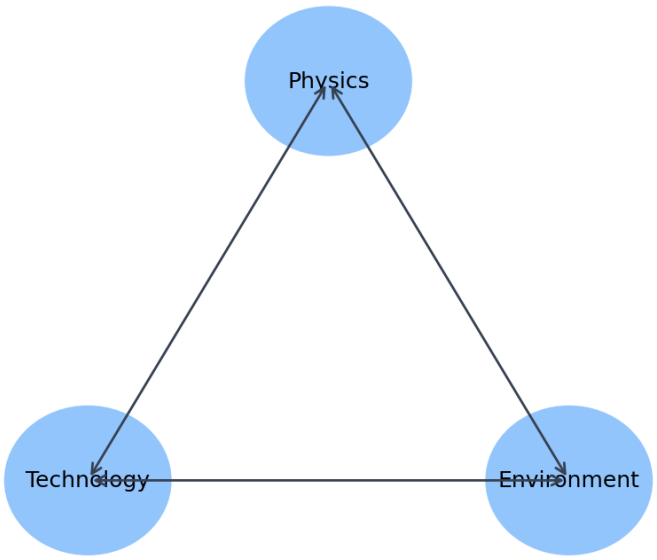
§ 42. Physics and scientific-technological progress

- Physics is a fundamental science whose discoveries underlie most modern technologies: from electricity and radio communication to computers, lasers, and nuclear power.
- Physics develops in a spiral: new experimental facts demand new theories (e.g. the shift from classical mechanics to relativity and quantum mechanics), and new theories open the way to new technologies.
- Kazakhstani scientists and research centres (e.g. the Institute of Nuclear Physics in Almaty) contribute to physics and its practical applications in nuclear power, medicine, and materials science.

DEFINITIONS

Term	Definition
Scientific-technological progress	the continuous, mutually reinforcing development of science and technology.

DIAGRAM §42



KEY POINT
The history of physics shows that fundamental discoveries (e.g. Maxwell's electromagnetism or Planck's quantum theory) often find practical use decades after they are made — so fundamental research matters even without an immediate payoff.

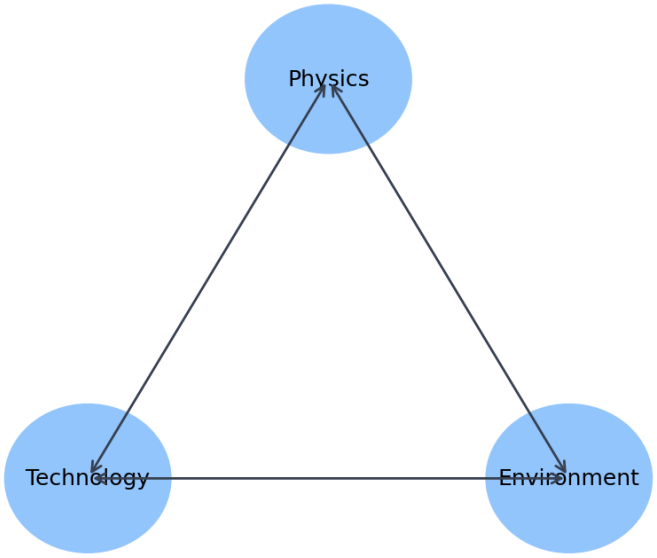
§ 43. The physical picture of the world

- The physical picture of the world is a system of views on the structure and laws of nature, formed on the basis of the physical theories of a given era; it has changed historically from a mechanical to an electromagnetic and then a quantum-relativistic picture.
- The modern physical picture of the world unites the theory of relativity (describing space, time, and gravity on large scales) and quantum mechanics (describing the behaviour of microparticles), and views the Universe as evolving, from the Big Bang to the present.
- Physics is closely tied to ecology and sustainable development: understanding physical laws is needed for rational energy use, reducing pollution, and finding new energy sources.

DEFINITIONS

Term	Definition
Physical picture of the world	a system of views on the structure and laws of nature, based on the physical theories of a given era.

DIAGRAM §43



KEY POINT
No physical picture of the world is final: it develops as new experimental data accumulates and theories improve, while remaining a key tool for explaining and predicting natural phenomena.